



Republic of Iraq
Middle Technical University
Technical Engineering College of Artificial Intelligence
Cybersecurity Engineering Technology Department

Computer Network Protocols

Chapter One

Network Overview

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1.1- Network Architecture

A set of layers and protocols is called the *network architecture*.

Protocol Hierarchies

- Networks are organized as layers to reduce design complexity.
- Each layer offers services to the higher layers. Between adjacent layers is an interface.

1.2- Network Protocols

Protocol:

A network protocol defines rules and conventions for communication between network devices. Network protocols include mechanisms for devices to identify and make connections with each other, as well as formatting rules that specify how data is packaged into sent and received messages.

Protocol Process:

1. The format or structure of the message
2. The process by which networking devices share information about pathways with other networks
3. How and when error and system messages are passed between devices
4. The setup and termination of data transfer sessions

1.3- Layering in Networked Computing

- OSI Model (Open System Interconnection)
- TCP/IP Model (Transmission Control Protocol/ Internet Protocol)

Why a layered model?

1. Breaks down communication into smaller, simpler parts.
2. Easier to teach communication process.
3. Allows different hardware and software to work together.
4. Reduces complexity.

1.4- The OSI model

The Open Systems Interconnection is the model developed by the International Standards Organization (ISO). The OSI model benefits are:

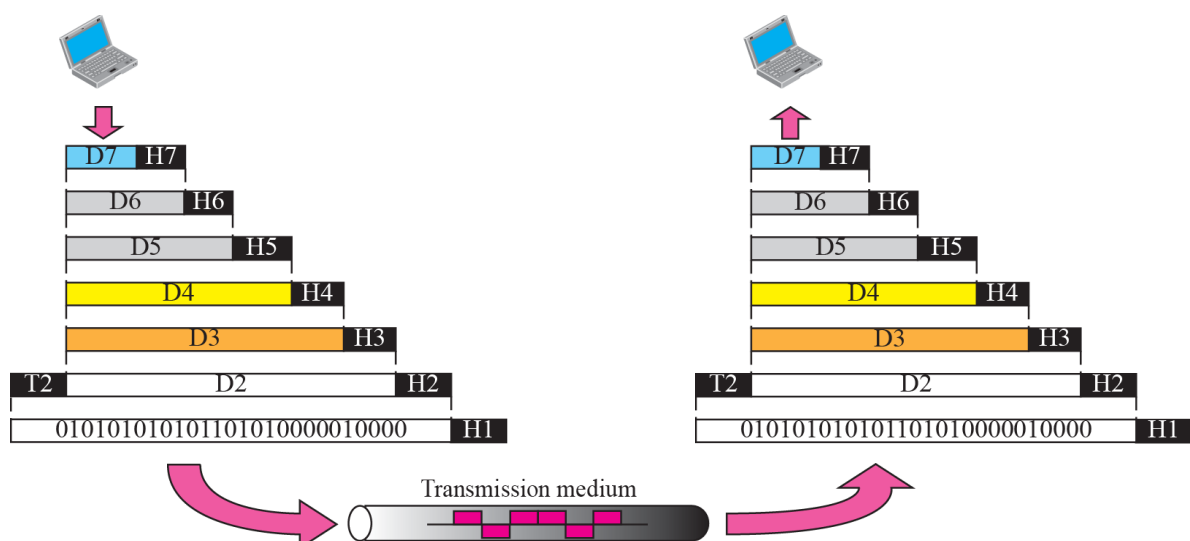
- Helps us understand how data gets from one user's computer to another.
- It aids to provide an organized structure for hardware and software developers to follow.

Why use a reference model?

- Serves as an outline of rules for how protocols can be used to allow communication between computers.
- Each layer has its own function and provides support to other layers.

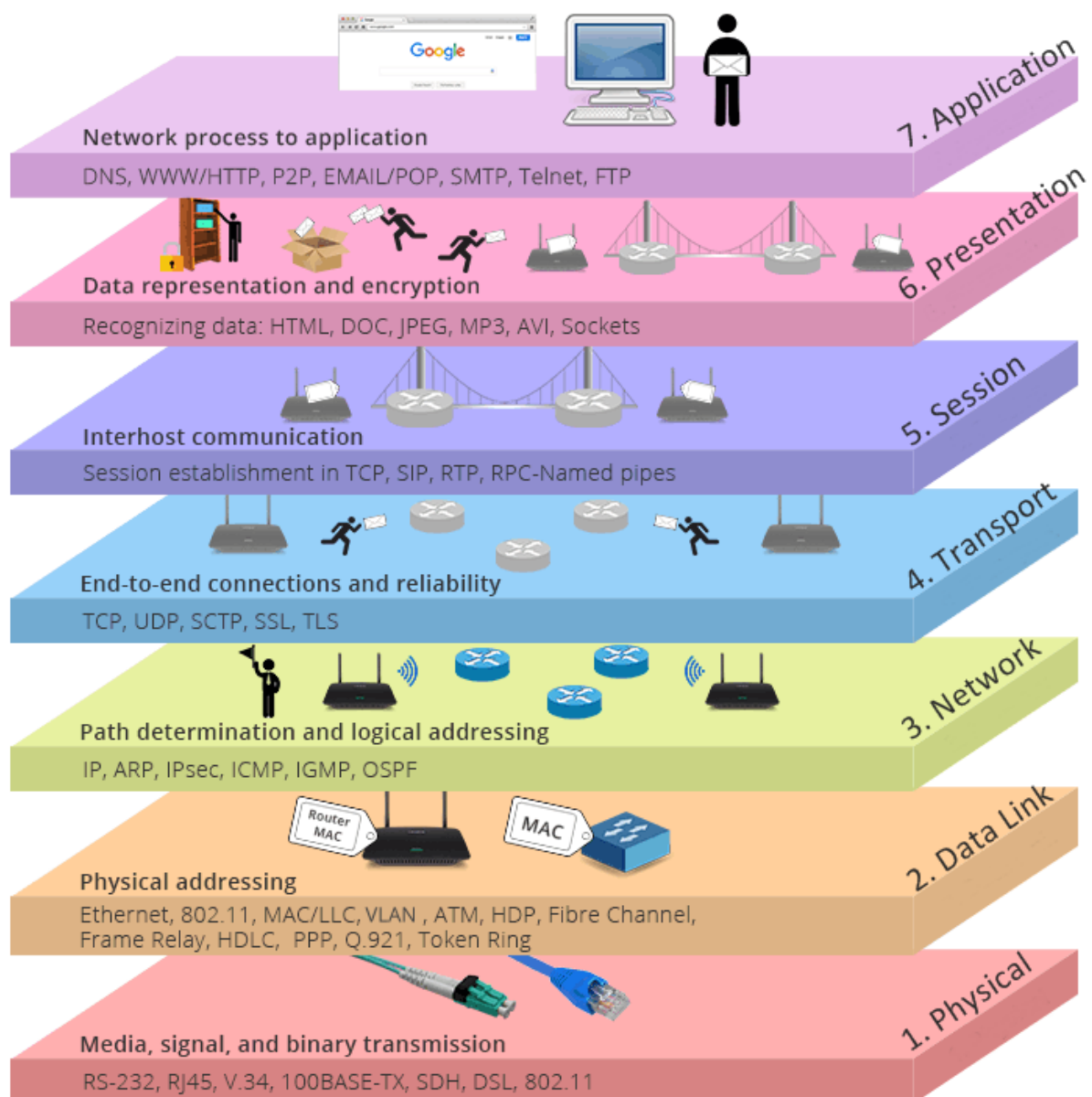
OSI model		
Layer	Name	Example protocols
7	Application Layer	HTTP, FTP, DNS, SNMP, Telnet
6	Presentation Layer	SSL, TLS
5	Session Layer	NetBIOS, PPTP
4	Transport Layer	TCP, UDP
3	Network Layer	IP, ARP, ICMP, IPsec
2	Data Link Layer	PPP, ATM, Ethernet
1	Physical Layer	Ethernet, USB, Bluetooth, IEEE802.11

At each layer, a header (H) can be added to the data unit. At layer 2, a trailer (T) may also be added.



Layer	Main Topics
Physical Layer	• Transmission mediums (transmit bits over medium) • Encoding • Modulation • Multiplexing • Topology • Provide mechanical and electrical specification • Repeaters & Hubs (multi-port repeater)
Data Link Layer	• Node to node (hop to hop) delivery • Frame format • Physical addressing (MAC Address) • Access control • Flow control • Error handling • Bridges & Switches (multi-port bridges)
Network Layer	• Internetworking • Controls the operation of the subnet. • Routing algorithms (Routing packets from source to destination) • Internet Protocol (IP) addressing (Logical addressing) • Routers
Transport Layer	• Connection-oriented and connectionless services • Provide reliable process to process message delivery & error recovery • Transmission Control Protocol (TCP) • User Datagram Protocol (UDP) • Provides additional Quality of Service. • Port address (service point address) • End-to-end flow control. • Segmentation and reassembly.
Session Layer	• Allows users on different machines to establish sessions (dialogue) between them. • managing dialogue control. • Session management. • Synchronization. • Graceful close

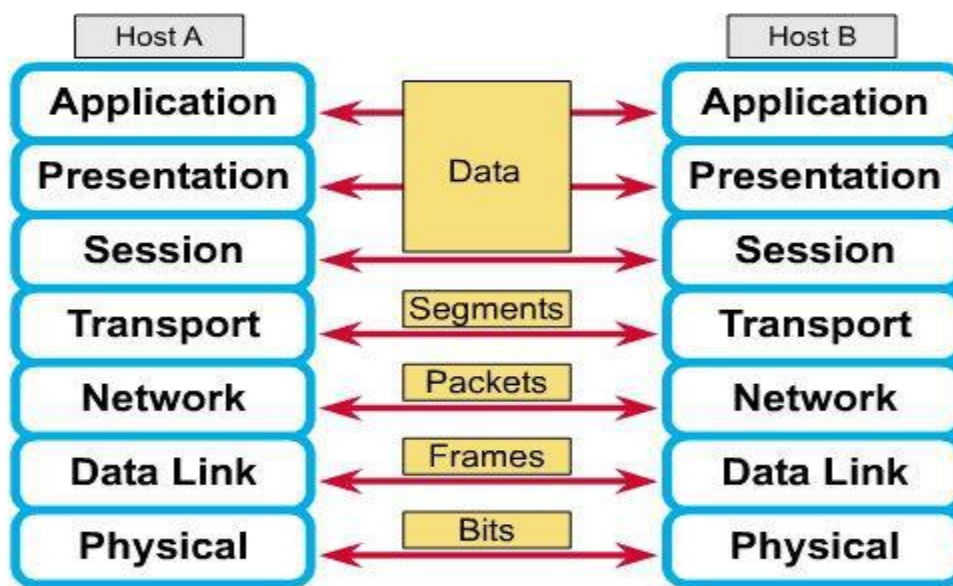
Presentation Layer	• Concerned with the syntax and semantics of the information. • Data Translation to preserve its meaning. • Data compression. • Data encryption. • Data Security
Application Layer	• Provides protocols that are commonly needed. • To allow access to network resource • (FTP), (HTTP), (SMTP), (SNMP),(NFS),(Telnet)



1.5- Data Encapsulation

Each layer contains a Protocol Data Unit (PDU)

1. PDU's are used for peer-to-peer contact between corresponding layers.
2. Data is handled by the top three layers, then Segmented by the Transport layer.
3. The Network layer places segment into packets and the Data Link frames the packets for transmission.
4. Physical layer converts frame to bits and sends it out over the media.
5. The receiving computer reverses the process using the information contained in the PDU (headers of each layer).

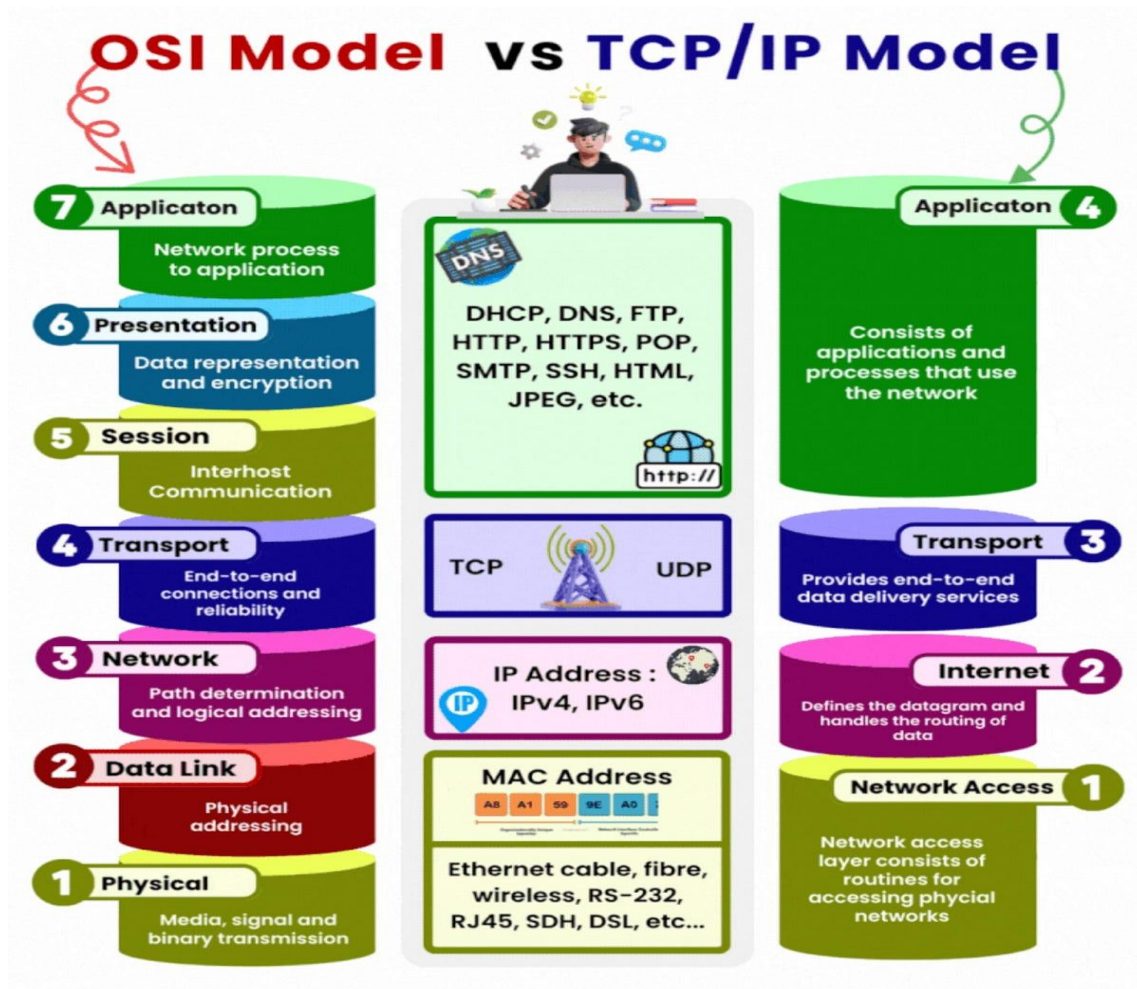


1.6- TCP/IP Model

The **TCP/IP protocol suite** was developed prior to the OSI model. Therefore, the layers in the TCP/IP protocol suite do not exactly match those in the OSI model. The original TCP/IP protocol suite was defined as having four layers:

- Layer 4: Application
- Layer 3: Transport
- Layer 2: Internet
- Layer 1: Network access

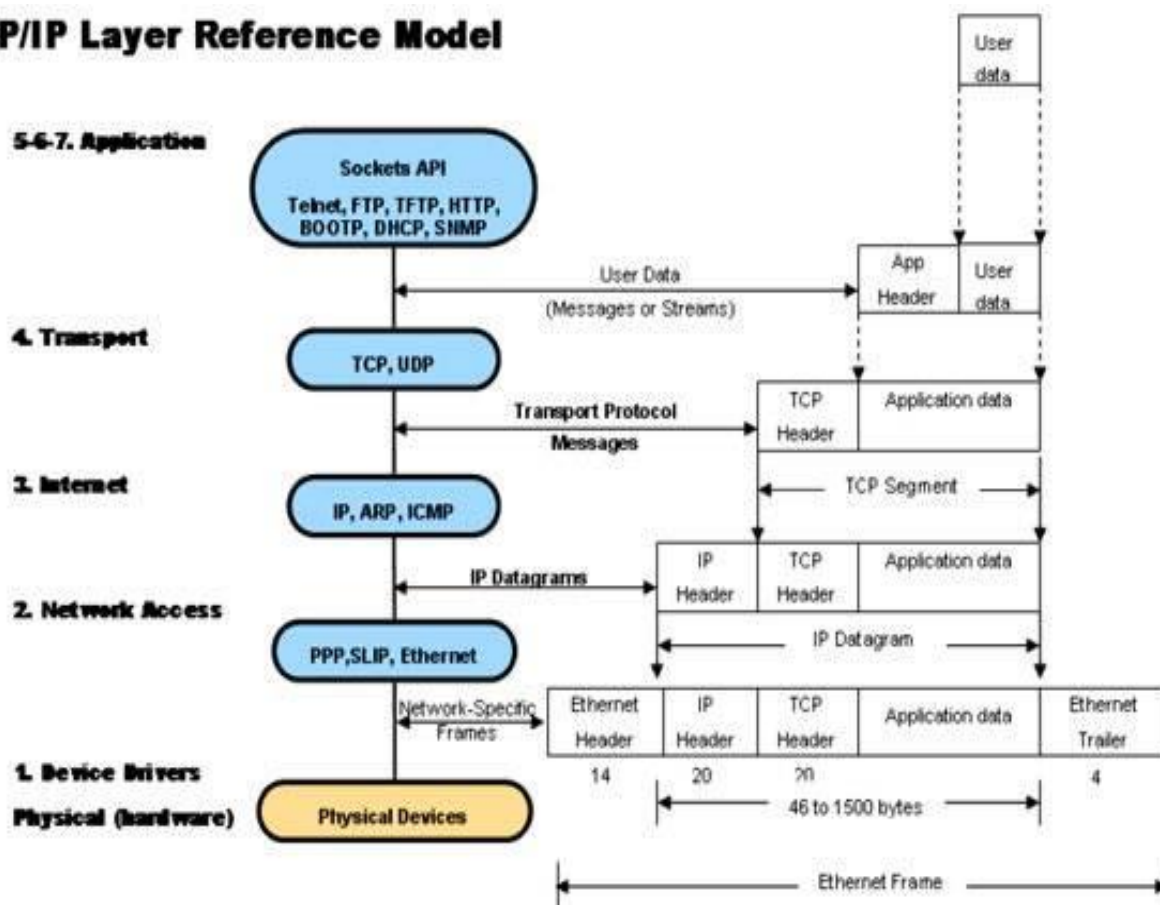
It is important to note that some of the layers in the TCP/IP model have the same name as layers in the OSI model. Do not confuse the layers of the two models.



Q: Compare between OSI and TCP/IP reference Models

Parameters	OSI Model	TCP/IP Model
No. of Layers	7 layers	4 Layers
Acronym	Open System Interconnection	Transmission Control Protocol/ Internet Protocol
Developed by	ISO (International Standard Organization)	DoD (Department of Defence)
Layer Separation	Has a separate session and presentation layers	Combine the session and presentation layers into the application layer
Protocol Implementation	Model was developed before the development of protocols	Protocols were developed then the model is developed
Model implementation	It is a theoretical model	It is the model around it the internet is developed
Reliable Delivery	It gives guarantee reliable packet delivery because it supports both connection oriented and connectionless communication in network layer	It does not give guarantee reliable packet delivery because it supports only connectionless communication in network layer

The TCP/IP Layer Reference Model



Q /Explain the delivery of data in Layered model?

Type of delivery	Layer	Shape of data	Type of addressing
End to End	Transport	Segment	Port (socket)
Source To Destination	Network	Packet	Logical (IP)
Node to Node	Data Link	Frame	Physical (MAC)

1.7- Categories of Networks

The category of a network can be determined according to its size into:

- *Local area networks (LANs).*
- *Metropolitan area networks (MANs).*
- *Wide area networks (WANs).*

1.7.1- Local Area Network (LAN)

A local area network is usually privately owned and links the devices in a single office building, or campus. A LAN can be as simple as two PCs and a printer in someone's home office or it can extend throughout a company and include voice, sound, and video peripherals.

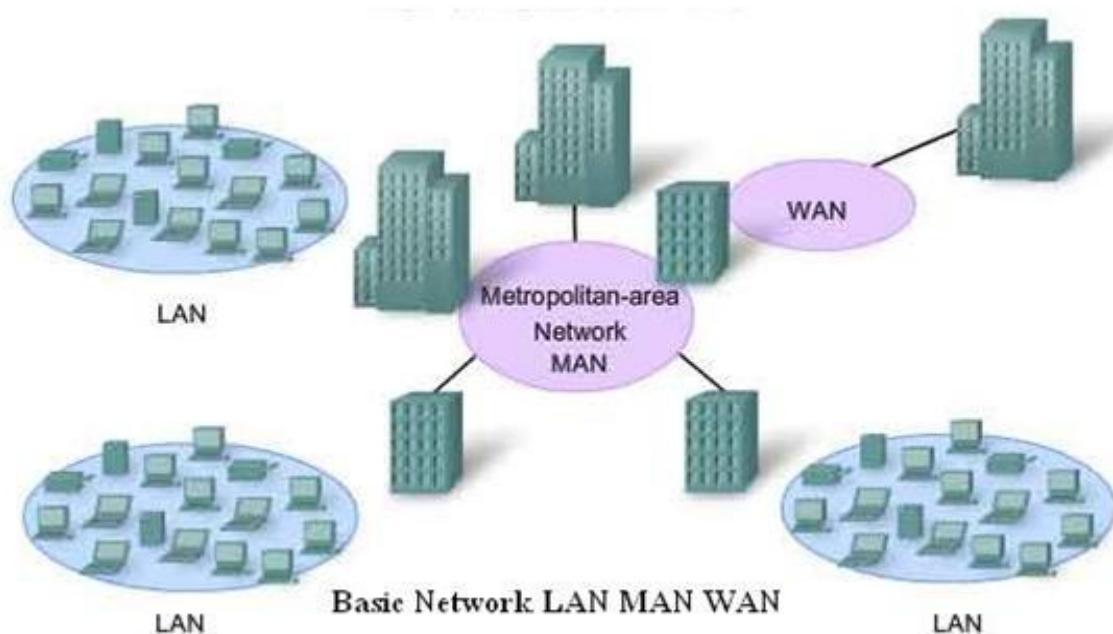
1.7.2- Metropolitan Area Network (MAN)

A metropolitan area network is designed to *extend over* an entire city. It may be a single network such as a cable television network or it may be a means of connecting a number of LANs into a larger network so that resources may be shared LAN-to-LAN as well as device-to-device. For example, a company can use a MAN to connect the LANs in all of its offices throughout a city.

A MAN may be wholly owned and operated by a private company, or it may be a service provided by a public company.

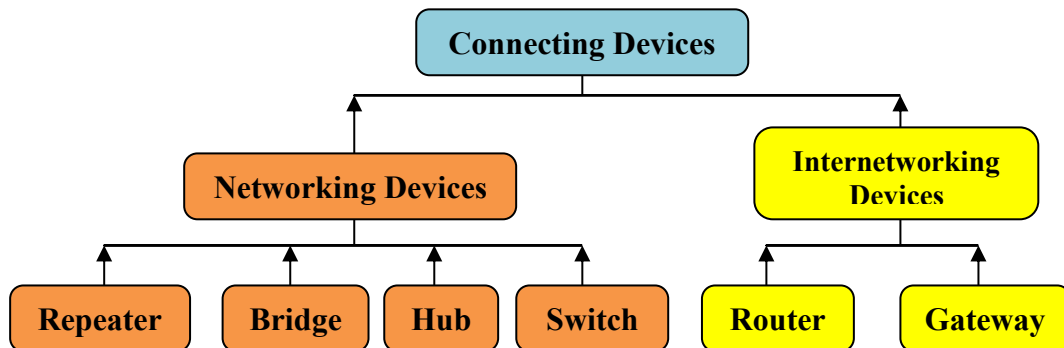
1.7.3- Wide Area Network (WAN)

A wide area network provides *long-distance* transmission of data; voice, image, and video information over large geographical areas that may comprise a country, a continent, or even the whole world. In contrast to LANs (which depend on their own hardware for transmission), WANs may utilize public, leased, or private communication devices, usually in combinations, and can therefore span an unlimited number of miles. A WAN that is wholly owned and used by a single company is often referred to as an *enterprise network*.

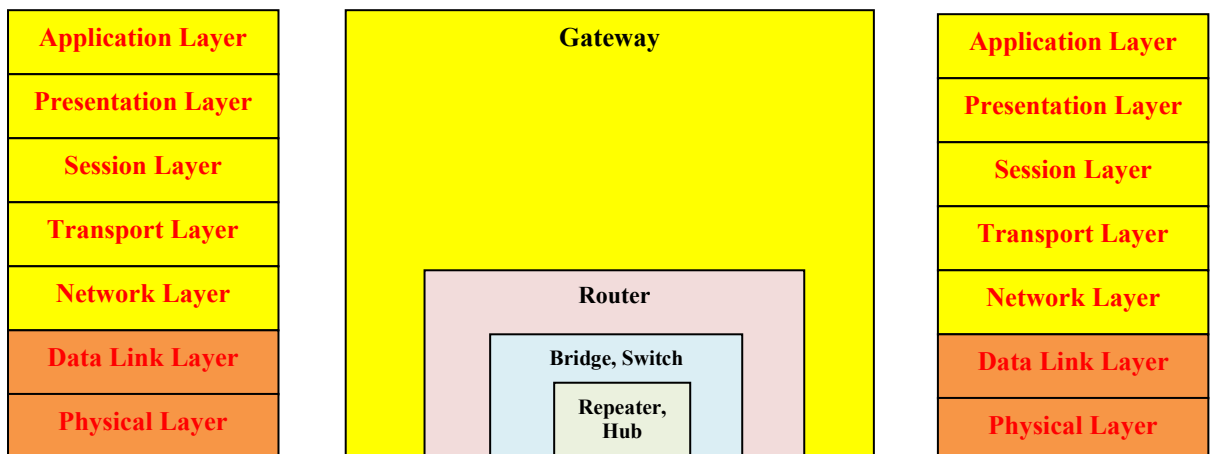


1.8- Networking and Internetworks

To connect a network or multiple networks together there are different types of connecting devices as illustrated in figure below:

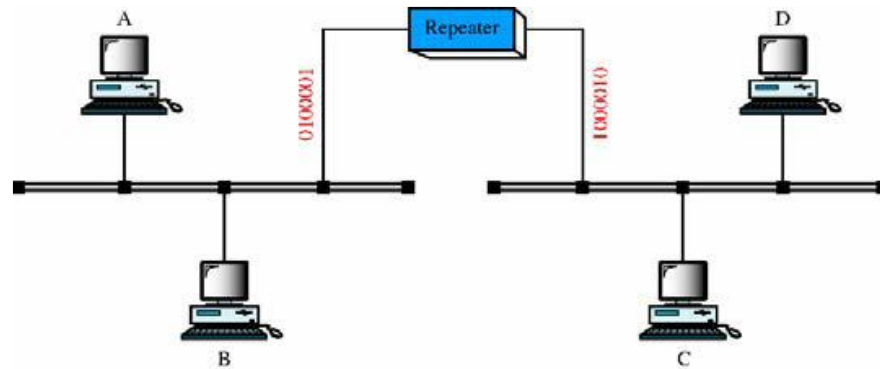


Each of these devices operates at different layer of the OSI model as shown in figure below:



1.8.1- Repeater:

Devices used to extend the network cable length beyond the limit of the specified cable. The purpose of a repeater is to **regenerate and retime network signals** at the bit level to allow them to travel a longer distance on the media. Repeaters are classified as **Layer 1 devices** in the OSI model, because they act only on the bit level and look at no other information.

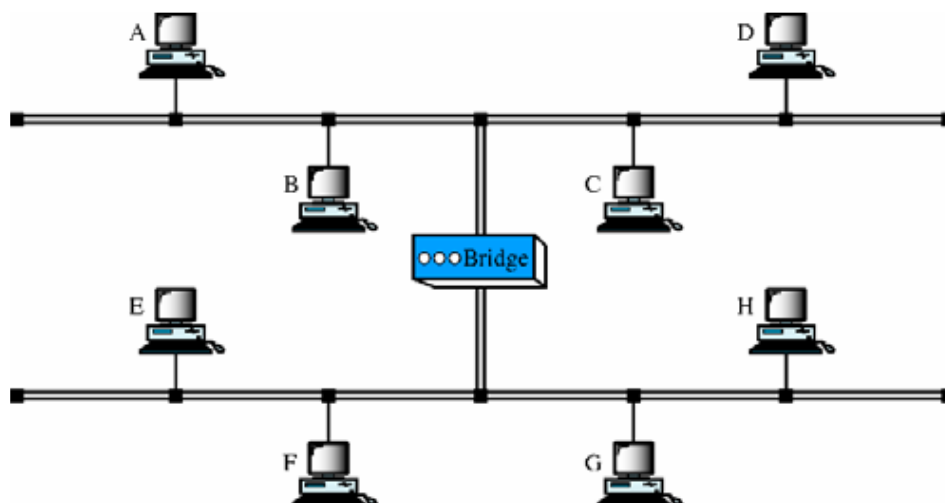


1.8.2- Hubs

The purpose of a hub is to **regenerate and retime network signals**. This is done at the bit level to a large number of hosts (e.g. 4, 8, or even 24) using a process known as concentration. You will notice that this definition is very similar to the repeaters, which is why a hub is also known as *a multi-port repeater*. Hubs are considered **Layer 1 devices** because they only regenerate the signal and broadcast it out all of their ports (network connections).

1.8.3- Bridges

A bridge is a **Layer 2 device** designed to connect two LAN segments. The purpose of a bridge is to filter traffic on a LAN, to keep local traffic local, yet allow connectivity to other parts (segments) of the LAN for traffic that has been directed there.



1.8.4 Switches

A switch is a **Layer 2 device** just as a bridge is. In fact, a switch is called a **multi-port bridge**, just like a hub is called a multi-port repeater. The difference between the hub and switch is that switches make decisions based on MAC addresses and hubs don't make decisions at all. Because of the decisions that switch make, they make a LAN much more efficient. They do this by "switching" data only out the port to which the proper host is connected. In contrast, a hub will send the data out all of its ports so that all of the hosts have to see and process (accept or reject) all of the data.

1.8.5- Routers

A router is a **Layer 3 device**. Working at Layer 3 allows the router to make decisions based on groups of network addresses as opposed to individual Layer 2 MAC addresses. Routers can also connect different Layer 2 technologies, such as Ethernet, Token-ring, and FDDI. However, because of their ability to route packets based on Layer 3 information, routers have become the backbone of the Internet, running the IP protocol. The purpose of a router is to examine incoming packets (Layer 3 data), choose the best path for them through the network, and then switch them to the proper outgoing port. A router is a type of internetworking device that passes data packets between networks, based on Layer 3 addresses. A router has the ability to make intelligent decisions regarding the best path for delivery of data on the network.

1.8.6- Gateway

A gateway is a **layer 7 device** and can be hardware or software, it works as a network connectivity device that connects two different configuration networks. Gateways are also known as protocol converters, because they play an important role in converting protocols supported by traffic on different networks. As a result, it allows smooth communication between two networks. It works as the entry-exit point for a network because all traffic that passes across the networks must pass through the gateway. A gateway monitors and controls all the incoming and outgoing network traffic.



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Chapter Two

The Internet Protocol (IP) **(IPv4 – IPv6)**

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2.1- Introduction

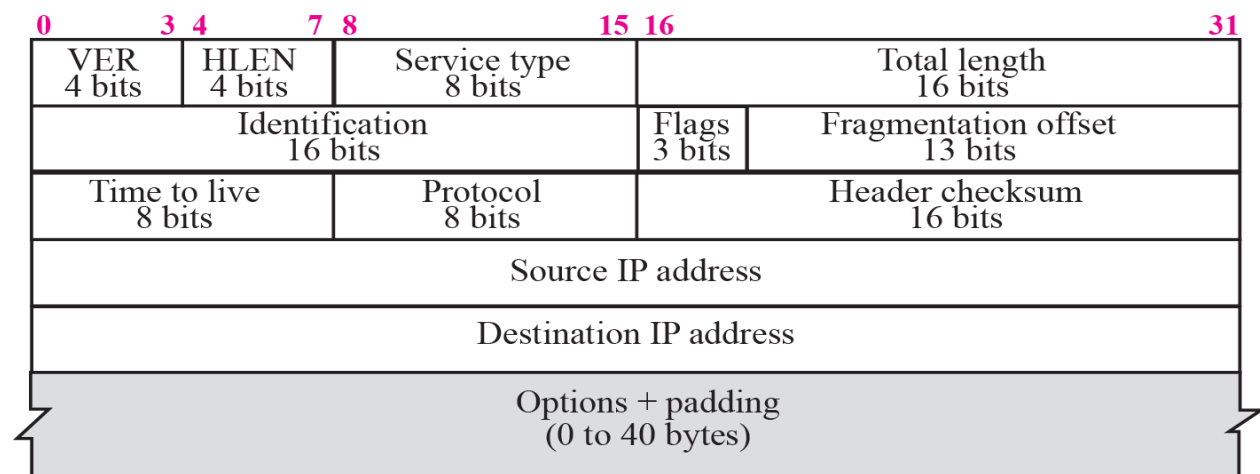
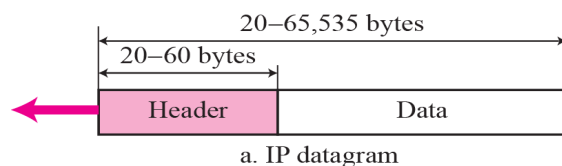
Communication at the network layer is source to destination; a computer somewhere in the world needs to communicate with another computer somewhere else in the world. Usually, computers communicate through the Internet. The packet transmitted by the sending computer may pass through several LANs or WANs before reaching the destination computer.

2.2- IPv4 Addressing

In the most widely installed level of the Internet Protocol (IP) today, an IPv4 (IP version 4) addresses or simply IP addresses if there is no confusion. It is a **32-bit address** that *uniquely* and *universally* defines the connection of a device (for example, a computer or a router) to the Internet.

2.3- IPv4 Datagram Format

Packets in the network (internet) layer are called **datagrams**. A datagram is a variable-length packet consisting of two parts: *header and data*. The header is **20 to 60 bytes** in length and contains information essential to routing and delivery. It is customary in TCP/IP to show the header in 4-byte sections.



b. Header format

IPv4 Header Structure

The IPv4 header typically contains 14 fields, 13 of which are mandatory.

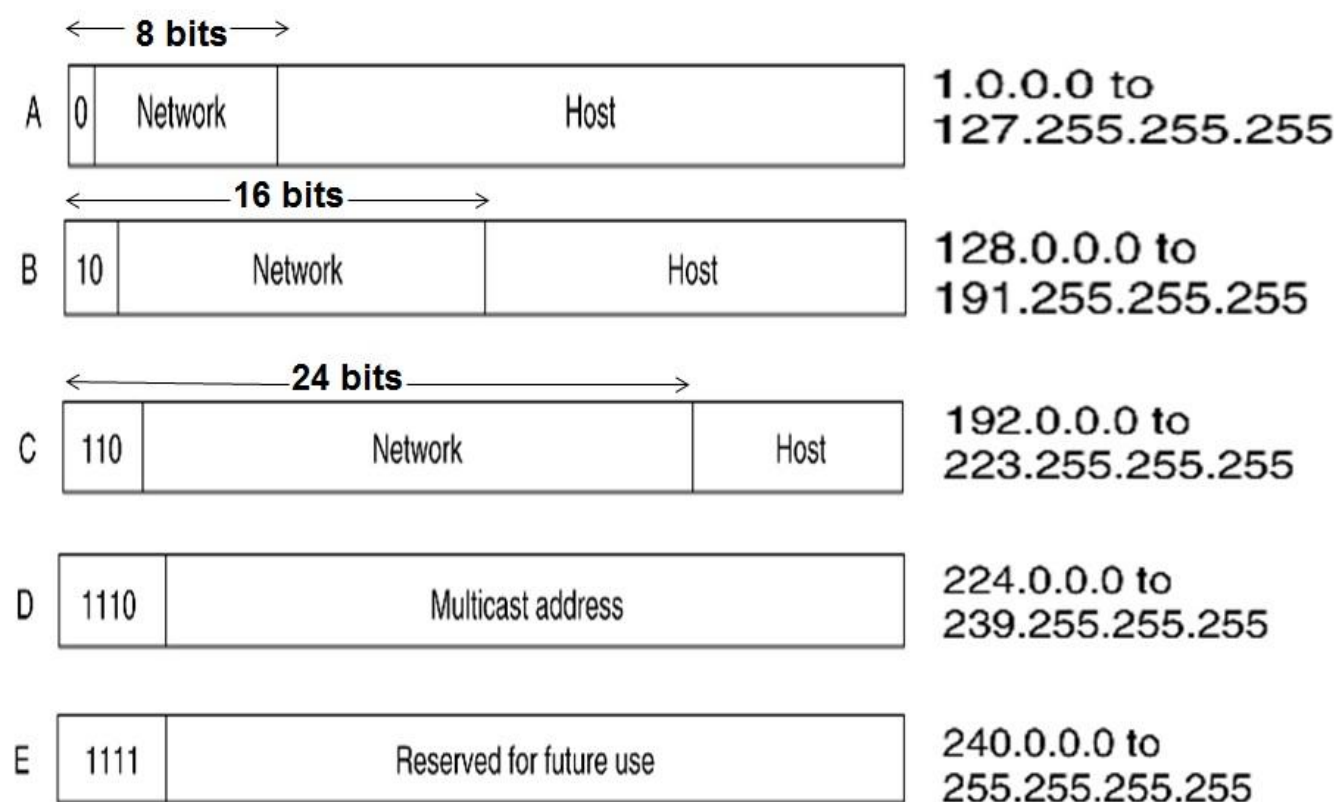
- **Version (4 bits):** Identifies the IP version. For IPv4, this is always 0100 (decimal 4).
- **Header Length (IHL) (4 bits):** Specifies the header size in 32-bit words. A value of 5 indicates the minimum 20-byte header, while 15 indicates the maximum 60-byte header.
- **Type of Service (ToS) / DSCP (8 bits):** Indicates the priority and quality of service (QoS) requirements for the datagram.
- **Total Length (16 bits):** Defines the entire size of the datagram (header + data) in bytes. The maximum possible size is 65,535 bytes.
- **Identification (16 bits):** Used to uniquely identify fragments of an original IP datagram to help with reassembly.
- **Flags (3 bits):** Controls fragmentation. The bits indicate if the packet can be fragmented and if there are more fragments to follow.
- **Fragment Offset (13 bits):** Indicates where in the original datagram a particular fragment belongs.
- **Time to Live (TTL) (8 bits):** A hop counter that prevents datagrams from looping indefinitely. Each router decrements this value by one; if it reaches zero, the datagram is discarded.
- **Protocol (8 bits):** Identifies the next-level protocol (e.g., TCP = 6, UDP = 17) to which the data should be passed.
- **Header Checksum (16 bits):** Used for error checking of the header only. It is recalculated at every hop because the TTL field changes.
- **Source IP Address (32 bits):** The 32-bit IP address of the sender.
- **Destination IP Address (32 bits):** The 32-bit IP address of the intended recipient.
- **Options (Variable):** Optional fields for testing, debugging, or security. If used, padding is added to ensure the header ends on a 32-bit boundary.

2.4- Classful IPv4 Addressing

IPv4 addressing, at its inception, used the concept of classes. This architecture is called classful addressing. Although this scheme is becoming obsolete, we briefly discuss it here to show the rationale behind classless addressing.

2.4.1- Classes

In classful addressing, the address space is divided into **five classes: A, B, C, D, and E**. Each class occupies some part of the address space.



2.4.2- Private IP addresses

These addresses can be used on a private network, but they are not routable through the internet. The range for the private IP addresses is given in table below

Range	Prefix
10.0.0.0 – 10.255.255.255	/ 8
172.16.0.0 – 172.31.255.255	/ 12
192.168.0.0 – 192.168.255.255	/ 16

The last column of Table above shows the mask in the form (/n) where n can be 8, 16, or 24 in classful addressing. This notation is also called *prefix notation* or **Classless Interdomain Routing (CIDR) notation**. The notation is used in classless addressing. We introduce it here because it can also be applied to classful addressing.

2.5- Classless Addressing

Classful allocation of address space often wasted many addresses, which exhausted the availability of IPv4 addresses. For example, a company that had a network with 260 hosts would need to be given a class B address with more than 65,000 addresses. The system that we currently use is referred to as classless addressing. With the classless system, address blocks appropriate to the number of hosts are assigned to companies or organizations without regard to the unicast class. This is can be achieved using many techniques such as:

- Subnetting
- Variable Length Subnet Mask (VLSM)
- Network Address Translation (NAT)

2.5.1- Subnetting

For the purpose of *reducing network traffics* and *simplifying network management*, and also *reducing the waste in IP addresses*; large network can be divided into smaller networks (subnetworks). Subnetting allows for creating multiple logical networks from a single address block.

We create the subnets by using one or more of the host bits as network bits. *This is done by extending the mask to borrow some of the bits from the host portion of the address to create additional network bits.* The more host bits used, the more subnets that can be defined. For each bit borrowed, we double the number of subnetworks available.

Notes:

- *The maximum number of borrowed bits is by leaving 2-bits in the HostID part of the address.*
- To determine the number of subnets or hosts use the following rules:

$$\text{Number of Subnets} = 2^{\text{number_of_borrowed_host_bits}}$$

$$\text{Usable Hosts Per Subnet} = (2^{\text{number_of_remaining_host_bits}} - 2)$$

Example: What are the number of subnets, hosts, and the valid host addresses given class C IP address 192.168.10.5 and a subnet mask 255.255.255.192 (192.168.10.5 / 26)

Solution:

IP address → 192.168.10.5 → 11000000 . 10101000 . 00001010 . 00000101

Subnet mask → 255.255.255.192 → 11111111 . 11111111 . 11111111 . 11000000

Network address → 192.168.10.0 → 11000000 . 10101000 . 00001010 . 00000000

Subnet mask → 255.255.255.192 → 11111111 . 11111111 . 11111111 . 11⁽⁰⁾000000

2 bits are borrowed

The number of subnets = 2^n . (where n = the number of bits borrowed)

$$= 2^2 = 4 \text{ subnets}$$

The number of hosts on each subnet = $2^m - 2$. (where m = the number of bits left for hosts).

$$= 2^6 - 2 = 62$$

shows that each of these subnets can have 62 hosts.

For each subnet, examine the last octet in binary. The values in these octets for the four networks are:

Subnet 1: **00000000** = 0

Subnet 2: **01000000** = 64

Subnet 3: **10000000** = 128

Subnet 4: **11000000** = 192

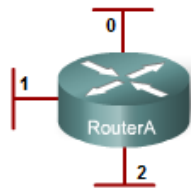
Or the subnet block size is = $2^{\text{number-of-host-bits}} = 2^6 = 64$ (*the magic number*)

Therefore, the subnets are = (0, 64, 128, 192)

Table below shows the valid host addresses and the subnet and broadcast addresses for each subnet.

Subnet	Subnet address	Host range	Broadcast address
1	192.168.10.0	192.168.10.1 → 192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65 → 192.168.10.126	192.168.10.127
3	192.168.10.128	192.168.10.129 → 192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193 → 192.168.10.254	192.168.10.255

Figure below illustrate the same process



-	192.168.1.0 (/24)	Address:	11000000.10101000.00000001.00000000
	255.255.255.0	Mask:	11111111.11111111.11111111.00000000
0	192.168.1.0 (/26)	Address:	11000000.10101000.00000001.00000000
	255.255.255.192	Mask:	11111111.11111111.11111111.11000000
1	192.168.1.64 (/26)	Address:	11000000.10101000.00000001.01000000
	255.255.255.192	Mask:	11111111.11111111.11111111.11000000
2	192.168.1.128 (/26)	Address:	11000000.10101000.00000001.10000000
	255.255.255.192	Mask:	11111111.11111111.11111111.11000000
3	192.168.1.192 (/26)	Address:	11000000.10101000.00000001.11000000
	255.255.255.192	Mask:	11111111.11111111.11111111.11000000

Two bits are borrowed to provide four subnets.

Unused address in this example.

A 1 in these positions in the mask means that these values are part of the network address.

More subnets are available, but fewer addresses are available per subnet.

2.5.2- VLSM

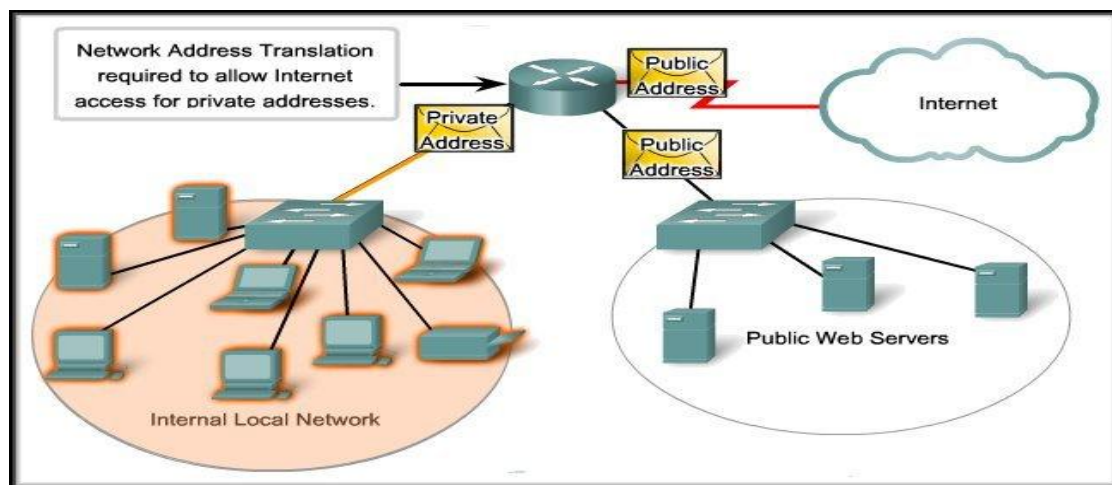
A serious limitation of using only a single subnet mask across a given network was that an organization is locked into a fixed-number of fixed-sized subnets. The solution to this problem is to use a **VLSM (variable length subnet mask)** enables a network number to be configured with different subnet masks on different interfaces.

2.5.3- Network Address Translation (NAT)

The number of home users and small businesses that want to use the Internet is ever increasing. The distribution of addresses through ISPs has created a new problem. If the business grows or the household needs a larger range, the ISP may not be able to grant the demand because the addresses before and after the range may have already been allocated to other networks. In most situations, however, only a portion of computers in a small network need access to the Internet simultaneously.

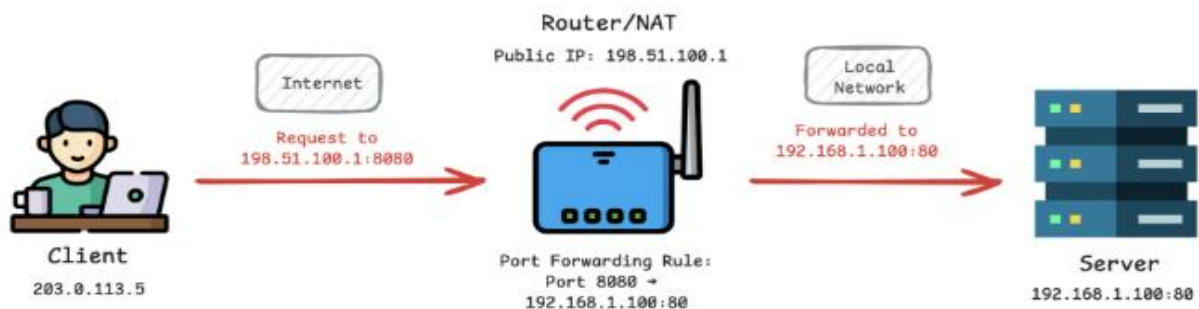
A quick solution to this problem is called **network address translation (NAT)**. NAT enables a user to have a large set of addresses internally and one address, or a small set of addresses, externally. The traffic inside can use the large set; the traffic outside,

the small set. The set of **private addresses** can be used **inside** the home or business network as shown in figure below.



2.5.4- Port Forwarding

Port Forwarding or **Port Mapping** is an application of network address translation (NAT) that redirects a communication request from one **address and port number combination** to another while the packets are traversing a network gateway, such as a router or firewall. This technique is most commonly used to make services on a host residing on a **protected or masqueraded** (internal) network available to hosts on the opposite side of the gateway (external network), by remapping the destination IP address and port number of the communication to an internal host

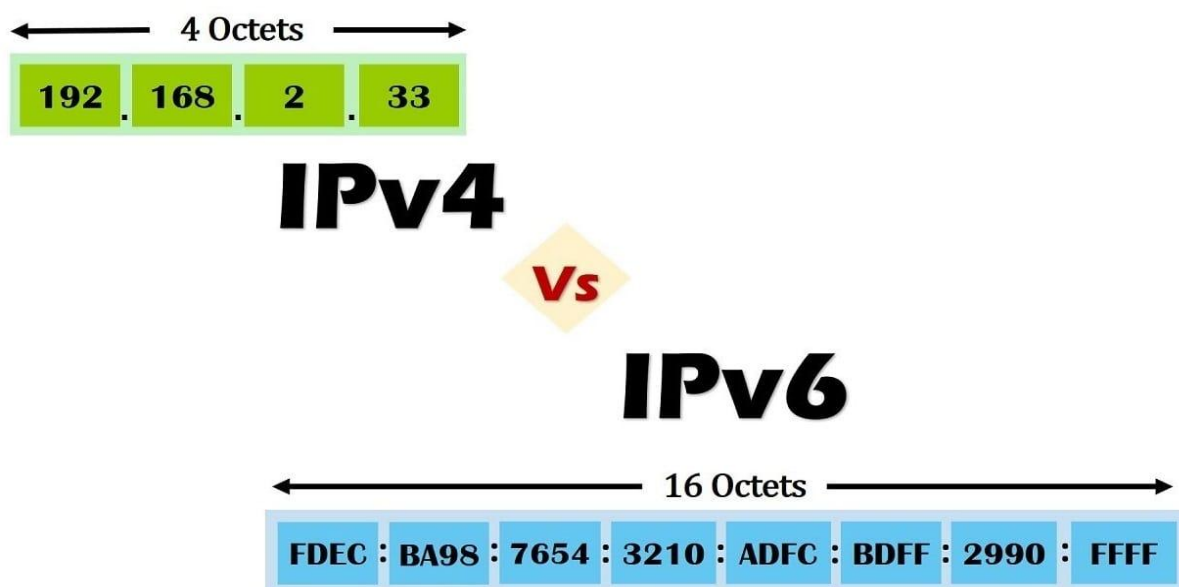


1. Client sends request to public IP (198.51.100.1) on port 8080
2. Router receives the request and checks port forwarding rules
3. Router forwards the request to internal server (192.168.1.100) on port 80
4. Server processes the request and sends response back through the same path

2.6- IPv6 Addressing

The need for more addresses, in addition to other concerns about the IP layer, motivated a new design of the IP layer called the new generation of IP or IPv6 (IP version 6). In this version, the Internet uses **128-bit addresses** that give much greater flexibility in address allocation. These addresses are referred to as IPv6 (IP version 6) addresses.

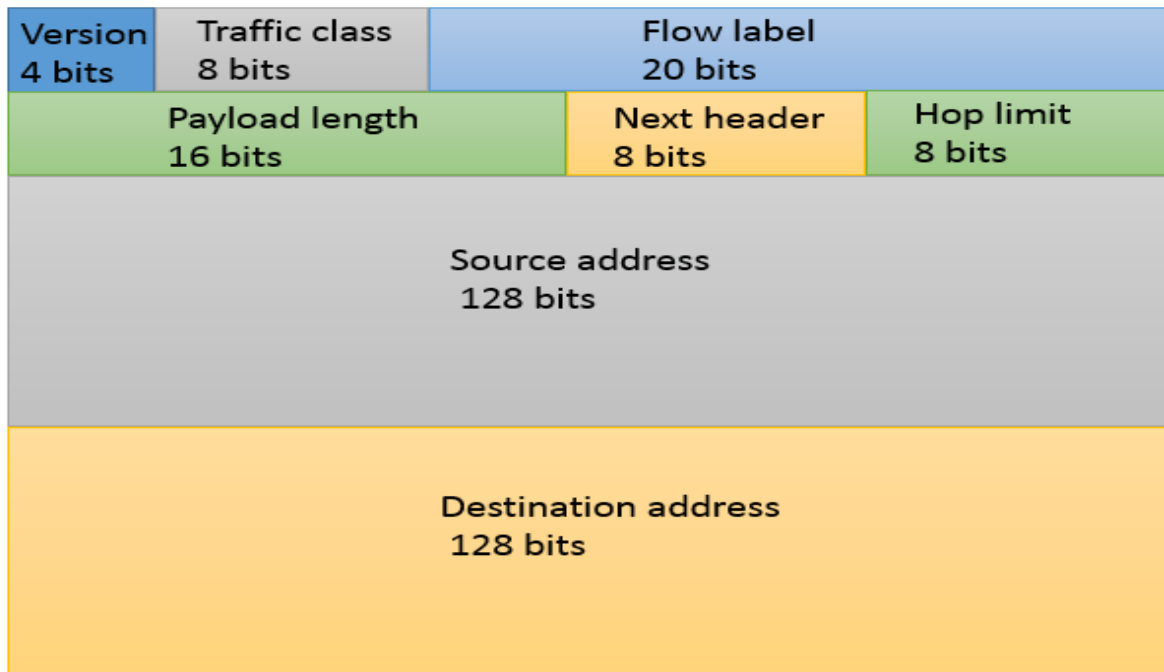
An IPv6 address consists of 16 bytes (octets); it is 128 bits long. To make addresses more readable, IPv6 specifies hexadecimal colon notation. In this notation, 128 bits is divided into eight sections, each 2 bytes in length. Two bytes in hexadecimal notation requires four hexadecimal digits. Therefore, the address consists of 32 hexadecimal digits, with every four digits separated by a colon, as shown below



2.6.1- IPv6 Features

The ability to scale networks for future demands requires a limitless supply of IP addresses and improved mobility; IPv6 combines expanded addressing with a more efficient and feature-rich header to meet these demands.

- header format helps speed processing/forwarding
- header changes to facilitate QoS



The main benefits of IPv6 include the following:

✚ **Larger address space:**

IPv6 addresses are 128 bits, this larger addressing space allows more support for addressing hierarchy levels, a much greater number of addressable nodes, and a simpler auto-configuration of addresses.

✚ **Globally unique IP addresses:**

Every node can have a unique global IPv6 address, which eliminates the need for NAT.

✚ **Header format efficiency:**

A simplified header with a fixed header size makes processing more efficient.

✚ **Improved privacy and security:**

IPsec is standard for IP network security, available for both IPv4 and IPv6.

✚ **Multicast instead of broadcast:**

With IPv6 the multicast communication substitutes the needs of broadcasting by using all node multicast

✚ **Stateless and stateful address configuration:**

IPv6 support besides the stateful addresses (DHCP) a stateless addresses autoconfiguration feature (SLAAC)

2.6.2- Compressed IPv6 Representation

There are two rules used to compress the IPv6 Representation

- **Rule 1 (Leading Zero Compression):** Eliminate the starting zero(s) from any hextex (zeros to the left).
- **Rule 2 (Successive Zeros Compression):** Eliminate group of successive zeros by one double-colon (::) but you can perform this only once in your IPv6 address.

Example: Given the following IPv6 address

(2001:1265:0000:0000:0AE4:0000:005B:06B0) used the rules to re-represent it in a compressed format.

Solution:

Original IP: 2001:1265:0000:0000:0AE4:0000:005B:06B0

After using Rule 1 the IP can be compressed into: 2001:1265:0:0:AE4:0:5B:6B0

After using Rule 2 the IP can be reduced into: 2001:1265::AE4:0:5B:6B0

2.6.3- IPv6 Three Address Types

In IPv6, there are three categories of IPv6 addresses:

- **Unicast**
- **Multicast**
- **Anycast**

Unicast Address:

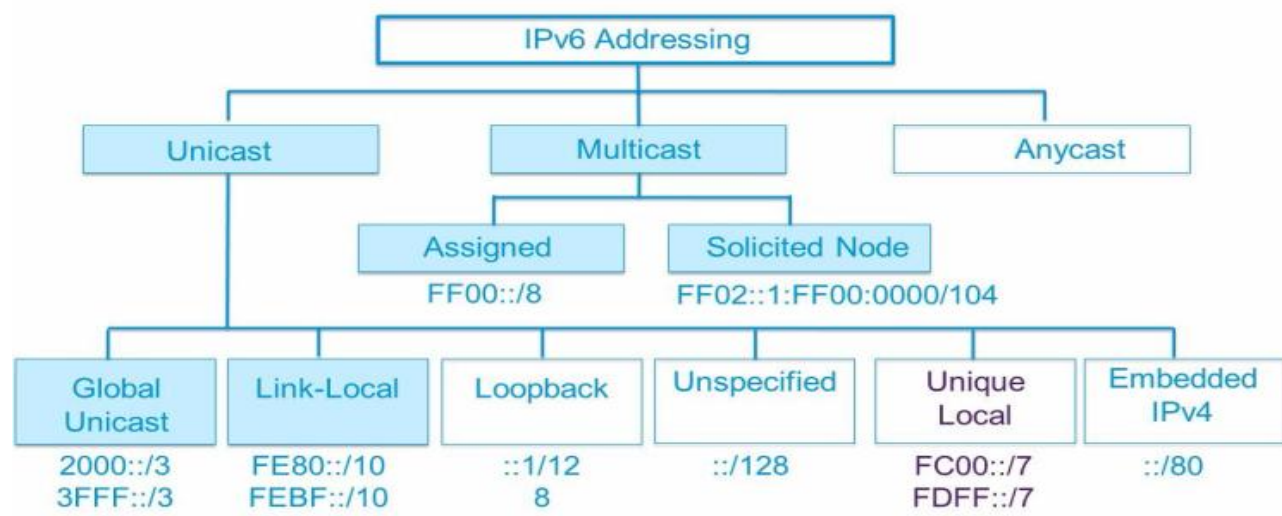
A unicast address defines a single interface (computer or router). The packet sent to a unicast address will be routed to the intended recipient.

Multicast Address:

A multicast address also defines a group of computers. In multicasting, each member of the group receives a copy.

Anycast Address:

An anycast address defines a group of computers that all share a single address. A packet with an anycast address is delivered to only one member of the group, the most reachable one.

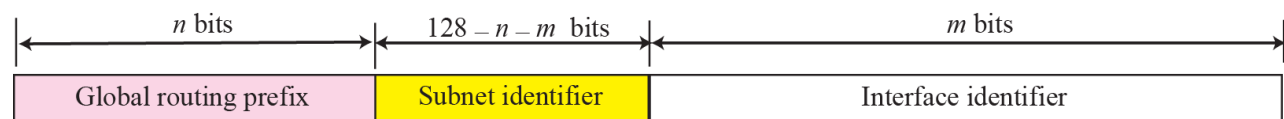


2.6.4- Three Levels of Hierarchy

An address in this block is divided into three parts:

- Global routing prefix
- Subnet identifier
- Interface identifier

as shown in Figure below:

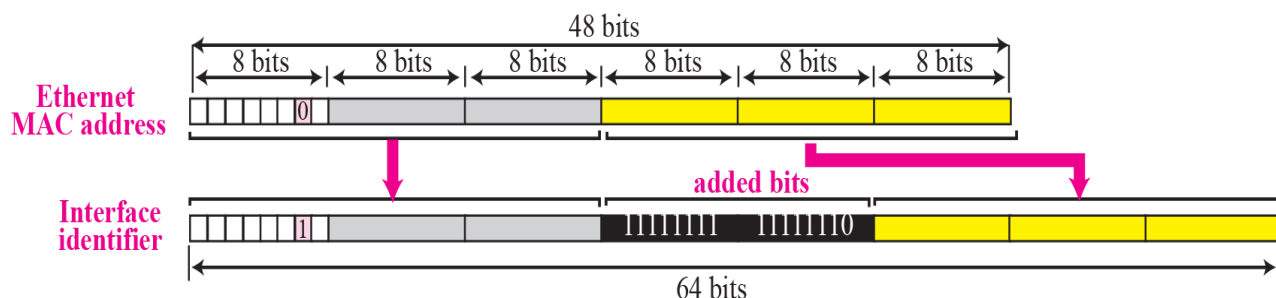


2.6.5- IPv6 Autoconfiguration

One of the interesting features of IPv6 addressing is the **autoconfiguration** of hosts. As we discussed in IPv4, the host and routers can be configured manually by the network manager. However, the Dynamic Host Configuration Protocol, DHCP, can be used to allocate an IPv4 address to a host that joins the network. In IPv6, DHCP protocol can still be used to allocate an IPv6 address to a host, but a host can also configure itself.

Interface Identifier

- The last 64 bits define the interface identifier.
- The interface identifier is similar to hostid in IPv4 addressing.
- The IPv6 addressing allows this opportunity. A physical address whose length is less than 64 bits can be embedded as the whole or part of the interface identifier, eliminating the mapping process. Mapping the 48-bit physical address defined by Ethernet to 64-bit extended unique identifier (EUI-64).



Example: Find the interface identifier if the Ethernet physical address is **(F5-A9-23-14-7A-D2)₁₆** using the format we defined for Ethernet addresses.

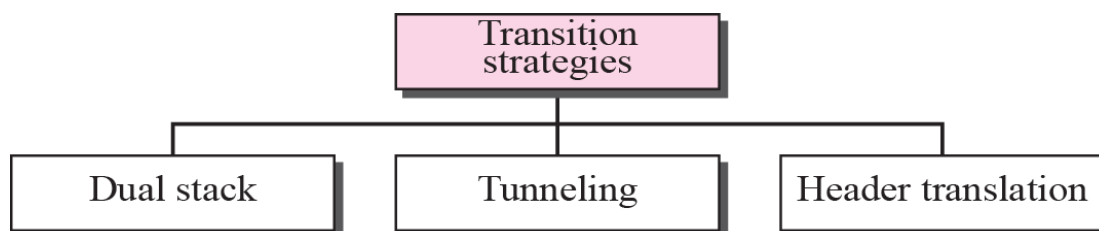
Solution:

We only need to change the seventh bit of the first octet from 0 to 1, insert two octets **(FFFE)₁₆** and change the format to colon hex notation.

The result is **F7A9:23FF:FE14:7AD2** in colon hex.

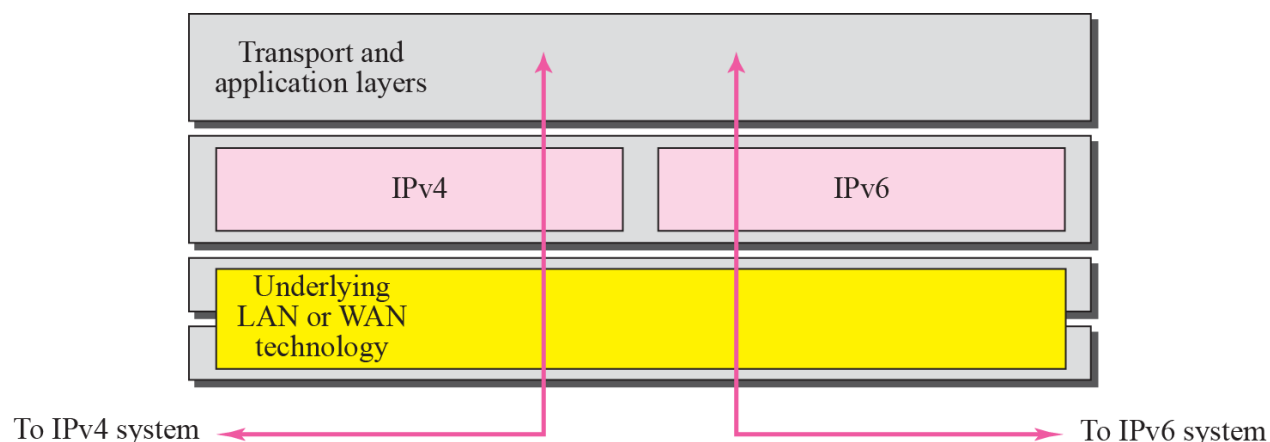
2.7- Transition from IPv4 to IPv6

Because of the huge number of systems on the Internet, the transition from IPv4 to IPv6 cannot happen suddenly. It will take a considerable amount of time before every system on the Internet can move from IPv4 to IPv6. The transition must be smooth to prevent any problems between IPv4 and IPv6 systems. Three strategies have been devised by the IETF to help the transition.



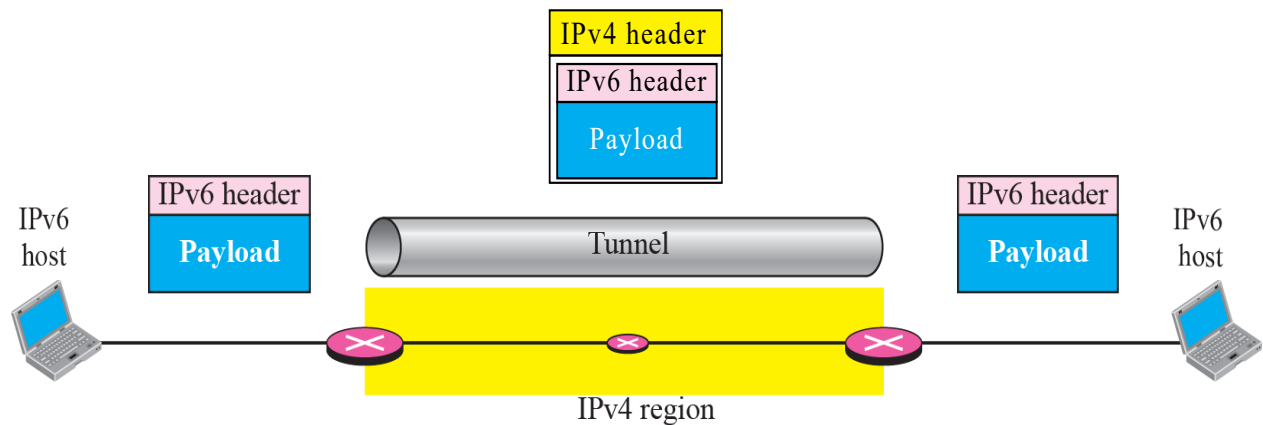
Dual Stack

It is recommended that all hosts, before migrating completely to version 6, have a dual stack of protocols. In other words, a station must run IPv4 and IPv6 simultaneously until all the Internet uses IPv6.



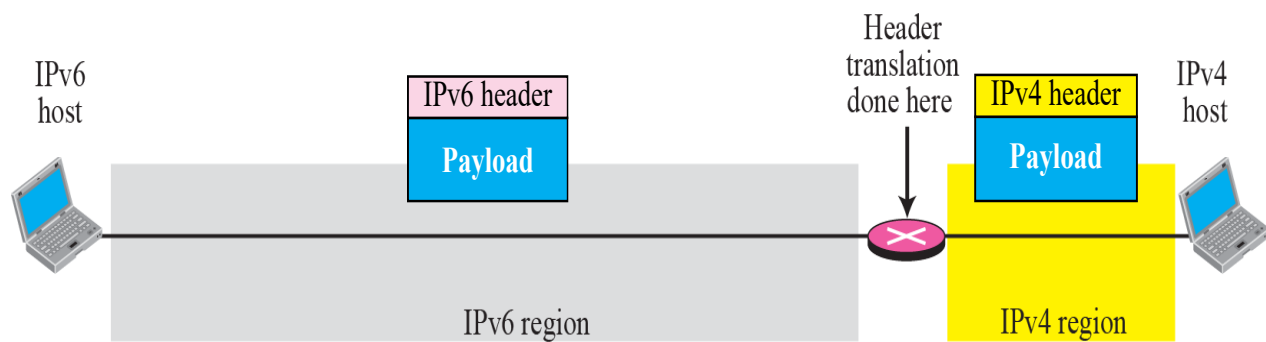
Tunneling

Tunneling is a strategy used when two computers using IPv6 want to communicate with each other, and the packet must pass through a region that uses IPv4. To make it clear that the IPv4 packet is carrying an IPv6 packet as data



Header Translation

Header translation is necessary when the majority of the Internet has moved to IPv6, but some systems still use IPv4. The sender wants to use IPv6, but the receiver does not understand IPv6. Tunneling does not work in this situation because the packet must be in the IPv4 format to be understood by the receiver. In this case, the header format must be totally changed through header translation (NAT64). The header of the IPv6 packet is converted to an IPv4 header.





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Chapter Three

ICMP, DHCP, IPSec & Forwarding Methods

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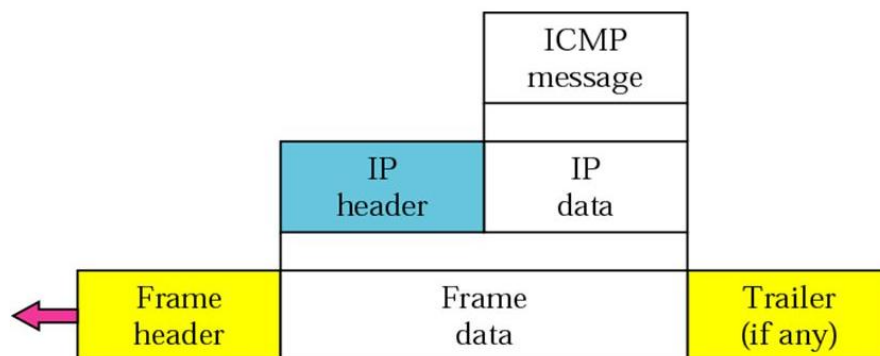
3.1- Internet Control Message Protocol (ICMP)

ICMP has been designed to compensate two deficiencies in IP protocol (Best effort delivery)

- ✓ Lack of error control
- ✓ Lack of assistance mechanism

3.1.1- ICMP Messages

ICMP itself is a network layer protocol. However, its messages are not passed directly to the data link layer as would be expected. Instead, the messages are first encapsulated inside IP datagrams before going to the lower layer.



ICMP messages are divided into two broad categories:

- **Error reporting messages:** report a problem that a router or a host (destination) may encounter during its processing to the IP packet.
- **Query messages:** help the host (source) or the network manager get specific information from a router or another host

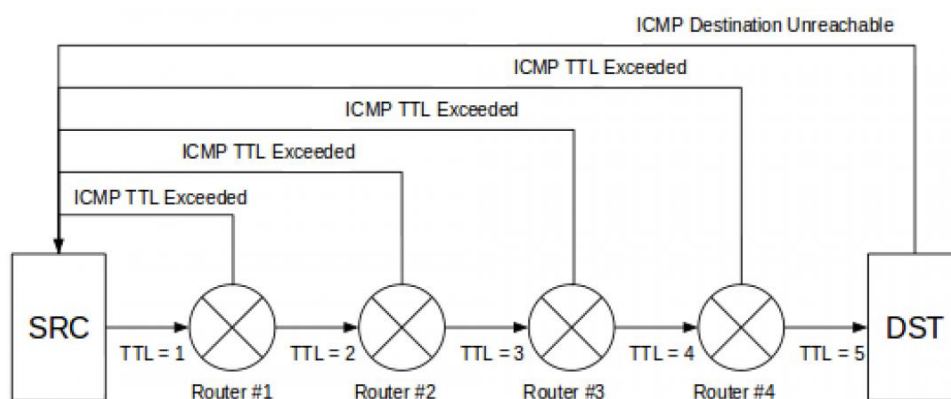
ICMP used by hosts & routers to communicate network-level information ICMP reports errors (unreachable host, router, port, or requested service is not available) and sends control message (echo request/reply (used by ping), ICMP does not attempt to make IP a reliable protocol. it simply attempts to report errors and provide feedback on the specific condition.

3.1.2- ICMP Applications

There are two simple and widely used applications that are based on ICMP:

Ping: The ping checks whether a host is alive & reachable or not. This is done by sending an ICMP Echo Request packet to the host and waiting for an ICMP Echo Reply from the host.

Traceroute: Traceroute is used to records the route through the Internet between your computer and a specified destination computer. It also calculates and displays the amount of time each hop took.



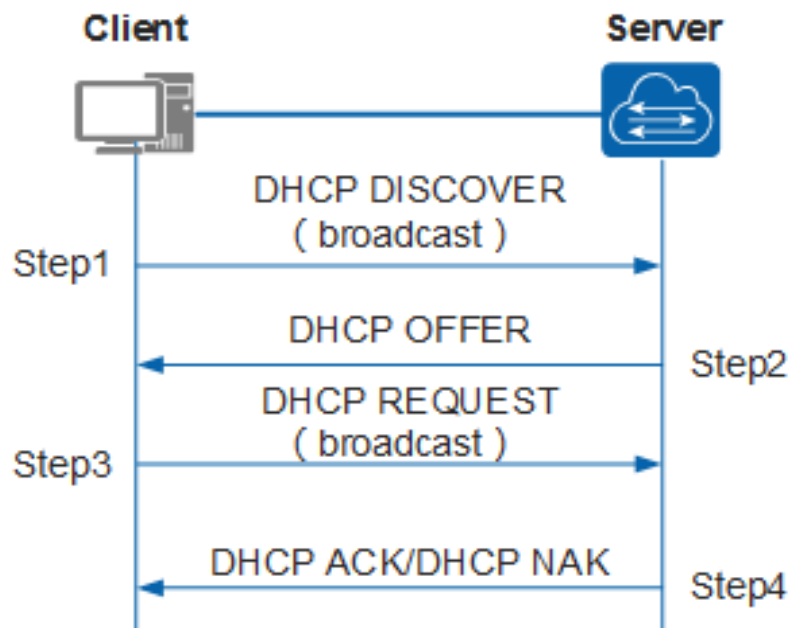
```
C:\Windows\system32\cmd.exe
C:\Users\admin>ping baidu.cn
Pinging baidu.cn [220.181.111.85] with 32 bytes of data:
Reply from 220.181.111.85: bytes=32 time=267ms TTL=52
Reply from 220.181.111.85: bytes=32 time=267ms TTL=52
Reply from 220.181.111.85: bytes=32 time=267ms TTL=52
Reply from 220.181.111.85: bytes=32 time=267ms TTL=52
Ping statistics for 220.181.111.85:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 267ms, Maximum = 267ms, Average = 267ms
C:\Users\admin>tracert baidu.cn
Tracing route to baidu.cn [220.181.111.85]
over a maximum of 30 hops:
  0  *          <1 ms    *
  1  <1 ms     <1 ms    <1 ms
  2  *          *        *
  3  *          *        *
  4  *          *        *
  5  135 ms    135 ms    135 ms
  6  *          *        *
  7  153 ms    151 ms    151 ms
  8  152 ms    151 ms    151 ms
  9  366 ms    366 ms    365 ms
 10  327 ms    330 ms    333 ms
 11  269 ms    267 ms    268 ms
 12  *          *        380 ms
 13  *          *        *
 14  408 ms    407 ms    411 ms
 15  *          *        *
 16  292 ms    292 ms    292 ms    220.181.111.85
Trace complete.
C:\Users\admin>
```

3.2- DHCP: Dynamic Host Configuration Protocol

Goal: allow a host to dynamically obtain its IP address from the network server when it joins the network.

DHCP overview: (pool operation)

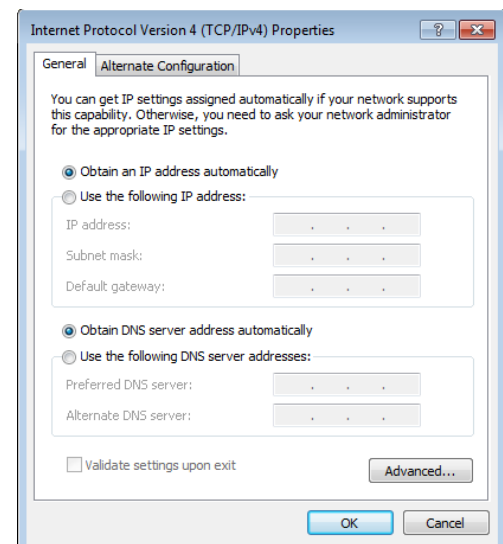
- host broadcasts “**DHCP discover**” msg
- DHCP server responds with “**DHCP offer**” msg
- host requests IP address: “**DHCP request**” msg
- DHCP server sends address: “**DHCP ack**” msg



DHCP: more than IP addresses

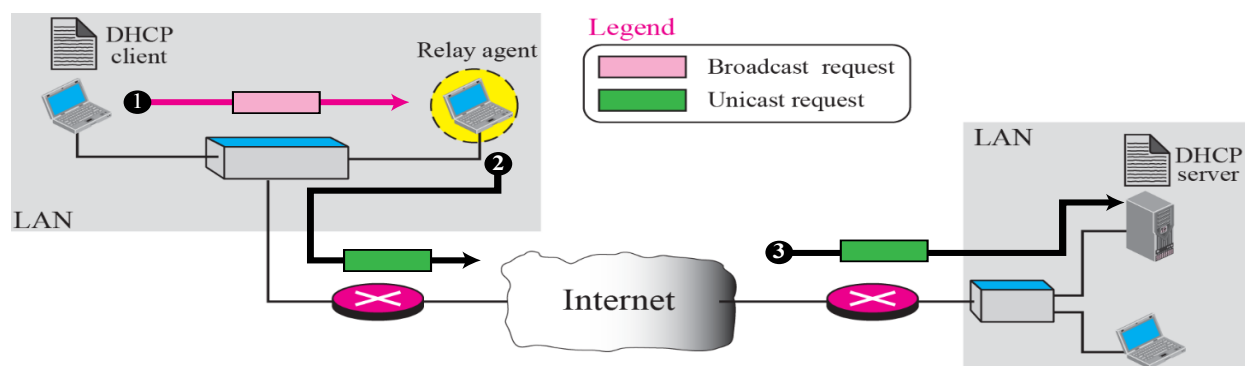
DHCP can return more than just an allocated IP address on the subnet:

- ✓ address of the first-hop router for client
- ✓ name and IP address of DNS server
- ✓ network mask



There is one problem in this method that must be solved. The DHCP request is broadcast because the client does not know the IP address of the server. A broadcast IP datagram cannot pass through any router.

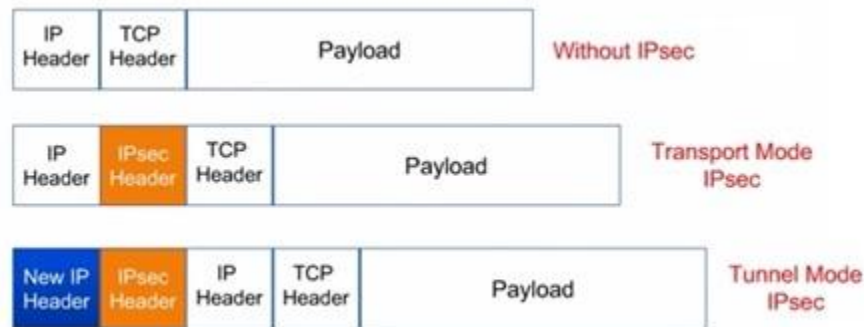
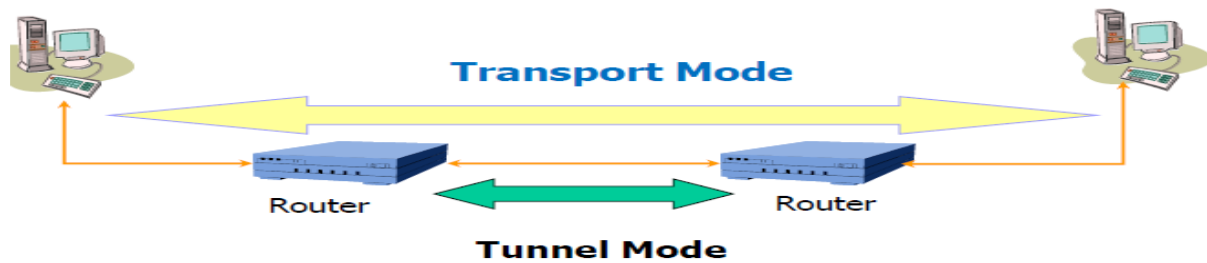
To solve the problem, there is a need for an intermediary. One of the hosts (or a router) can be used as a relay. The host in this case is called a relay agent.



3.3- IP Security (IPSec)

- IP Security (IPSec) is a collection of protocols designed by the Internet Engineering Task Force (IETF) to **provide security for a packet at the network level**.
- IPSec helps to create authenticated and confidential packets for the IP layer.
- IPSec operates in one of two different modes: the **transport mode** or the **tunnel mode**.

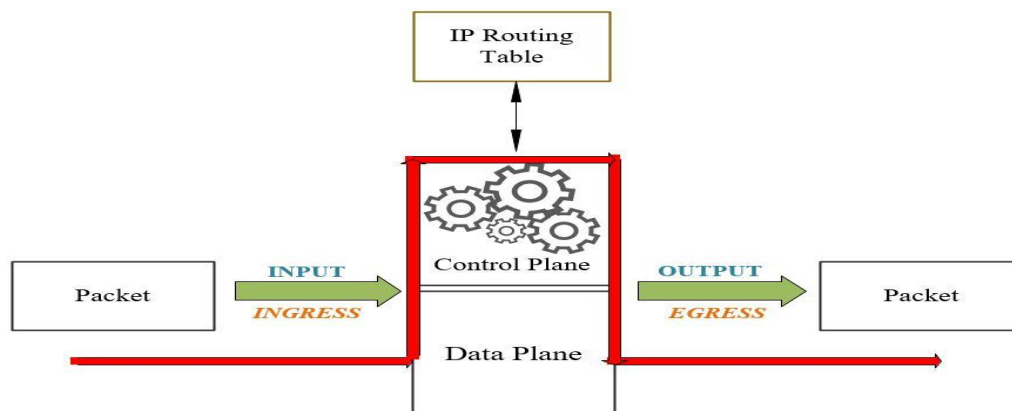
Transport Mode	Tunnel Mode
IPSec in the transport mode does not protect the IP header; it only protects the information coming from the transport layer.	IPSec in the tunnel mode protects the original IP header.
IP Header not Encrypted	IP Header Encrypted
Used for end-to-end communications (client and server)	Most commonly used between routers or ASA firewall (gateways)



IPsec Packet format transport mode vs. tunnel mod

3.4- Packet Forwarding

Packet forwarding is the process by which a network device, typically a router, receives a data packet on an input interface and transmits it to the appropriate output interface based on the packet's destination address. It is the local operation of moving a packet within a single node, distinct from routing, which involves building the global tables used for these forwarding decisions.



Core Forwarding Mechanism

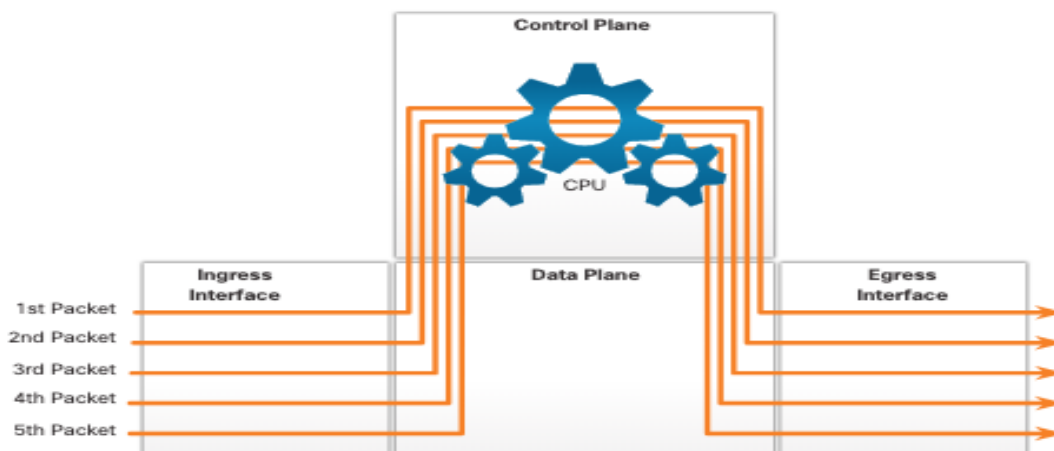
When an IP packet arrives at a router, it undergoes a series of standardized processing steps:

- **Validation:** The router verifies the IP header for correctness (e.g., checksum, version).
- **TTL Adjustment:** The Time-to-Live (TTL) field is decremented. If it reaches zero, the packet is discarded to prevent infinite loops.
- **Table Lookup:** The router extracts the destination IP address and performs a Longest Prefix Match (LPM) lookup in its forwarding table.
- **Next-Hop Determination:** The lookup identifies the next-hop neighbor or the exit interface.
- **Encapsulation & Transmission:** The packet is re-encapsulated with a new Layer 2 (MAC) header for the next link and placed in an output queue for transmission.

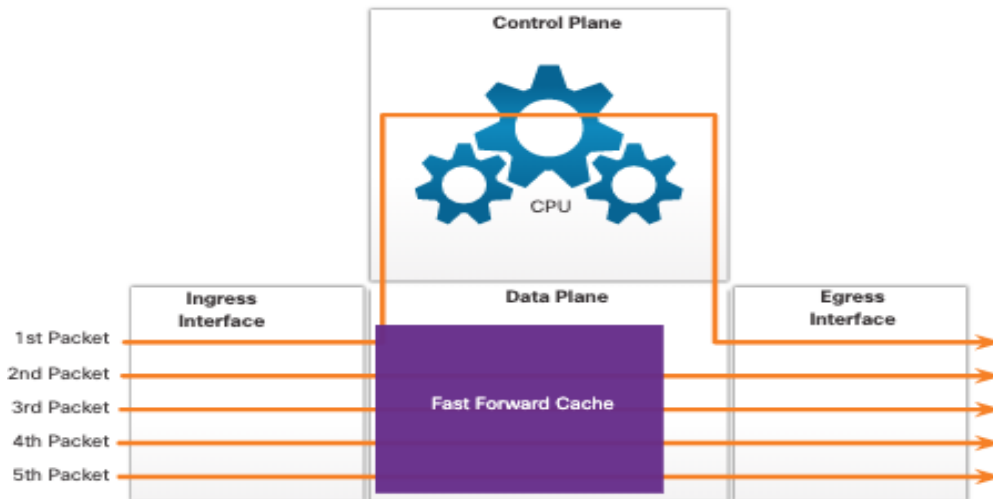
Major Implementation Methods

Different architectures prioritize speed or flexibility. Common mechanisms, notably in Cisco and other hardware routers, include:

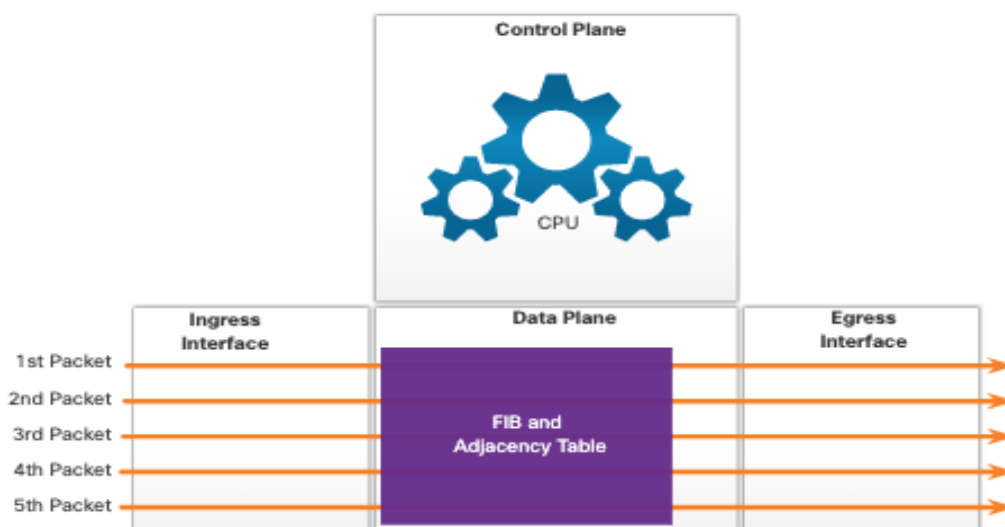
- ✚ **Process Switching:** The CPU processes every single packet individually. It is slow because it requires the control plane to handle every packet.



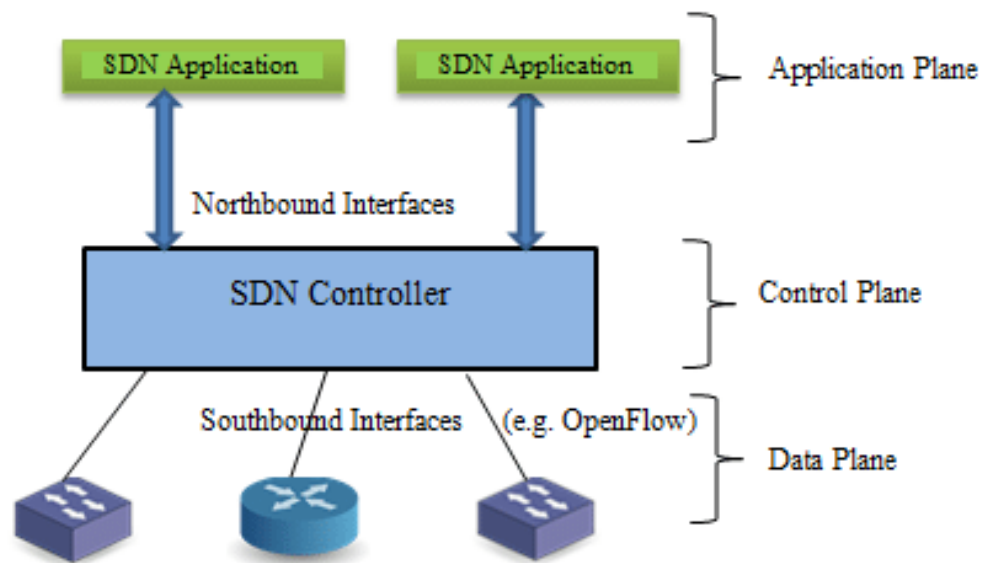
- ✚ **Fast Switching:** Only the first packet of a flow is processed by the CPU. The result is cached in a "**fast-switching cache**," allowing subsequent packets in the same flow to be forwarded without CPU intervention.



- ✚ **Cisco Express Forwarding (CEF):** The most advanced and default modern method. Its pre-computes all routing information into a **Forwarding Information Base (FIB)** and an **adjacency table** before packets arrive. This allows for line-rate forwarding entirely in hardware.



- ✚ **Software-Defined Networking (SDN):** Decouples the **control and data planes**. Forwarding is based on "**flow entries**" pushed to switches by a centralized controller using protocols like **OpenFlow**.



Software defined networking (SDN) is an approach to network management that enables dynamic, programmatically efficient network configuration to **improve network performance and monitoring**. It is a new way of managing computer networks that makes them easier and more flexible to control.

In traditional networks, the hardware (like routers and switches) decides how data moves through the network, but SDN changes this by moving the decision-making to a central software system. This is done by separating the **control plane** (which decides where traffic is sent) from the **data plane** (which moves packets to the selected destination).



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Chapter Four

Routing Algorithms & Protocols

Prepared By:

Assistant Prof. Dr. Mahmoud Shuker Mahmoud

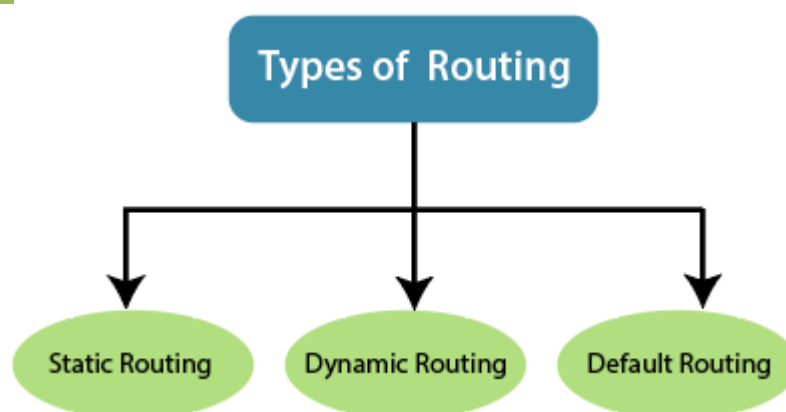
mahmoud.shukur@mtu.edu.iq

4.1- Routing Concept & Algorithms

ROUTING: is **forwarding** of packets from one network to another network choosing the **best path** from the routing table.

Routing table consist of only the *best routes* for every destination.

Types of Routing:



Static	• It is configured by Administrator manually • Need for destination network ID • It is secure and fast • Used for small organization which have network of 10-15 routers
Dynamic	• Means automatically routing • Dynamic routes mean that the router learns the paths of destinations by receiving periodic updates from other routers • Is automatically choose the best shortest path • Can be done by using routing protocol

Q: list the Advantages of dynamic over static routing?

- There is **no need** to know the destination networks
- **Updates** the topology changes **dynamically**
- Administrative work **reduced**
- Used for **large organizations**

4.2- Routing Algorithms

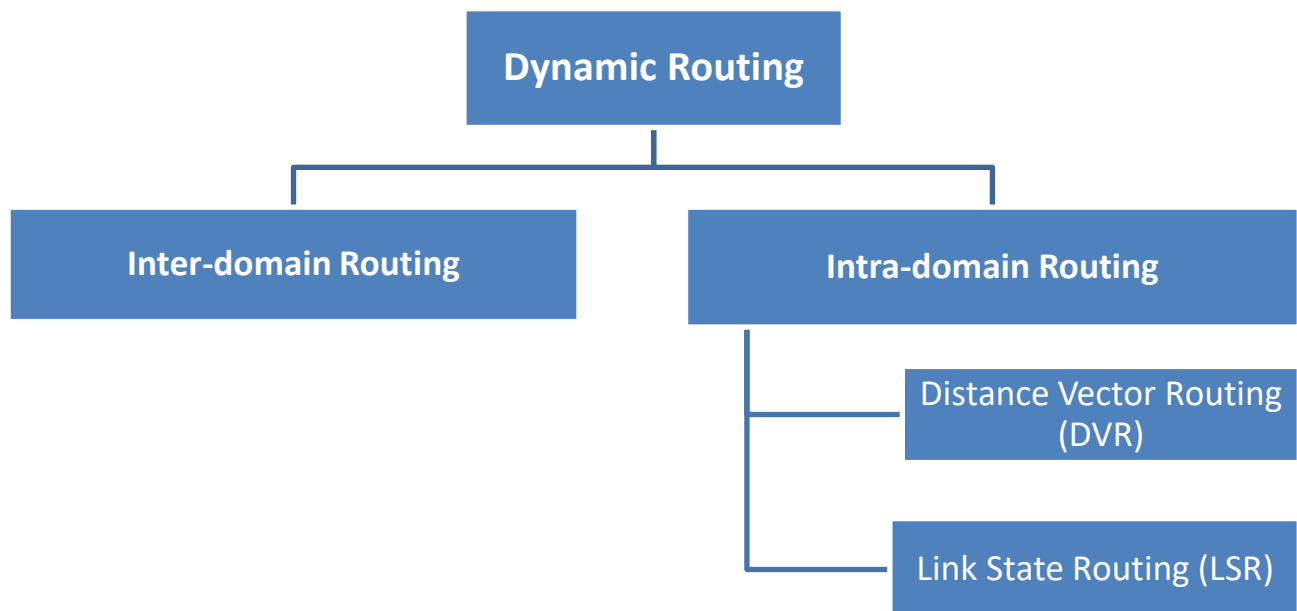
In the world of computers, any procedure can be illustrated through step by step, which is called an **algorithm**. Similarly, the routing algorithm is a step-by-step procedure for transferring data from one location to another over the network.

- The main function of the network layer is **routing packets from source to destination**.
- The **routing algorithm** is that part of the network layer software responsible for **deciding** which output line an incoming packet should be transmitted on.
- Routing algorithms can be grouped into two major classes: *non adaptive (Static)*, and *adaptive (Dynamic)*.

Dynamic Routing Algorithms

1- **Distance Vector Routing (DVR).**

2- **Link State Routing (LSR).**



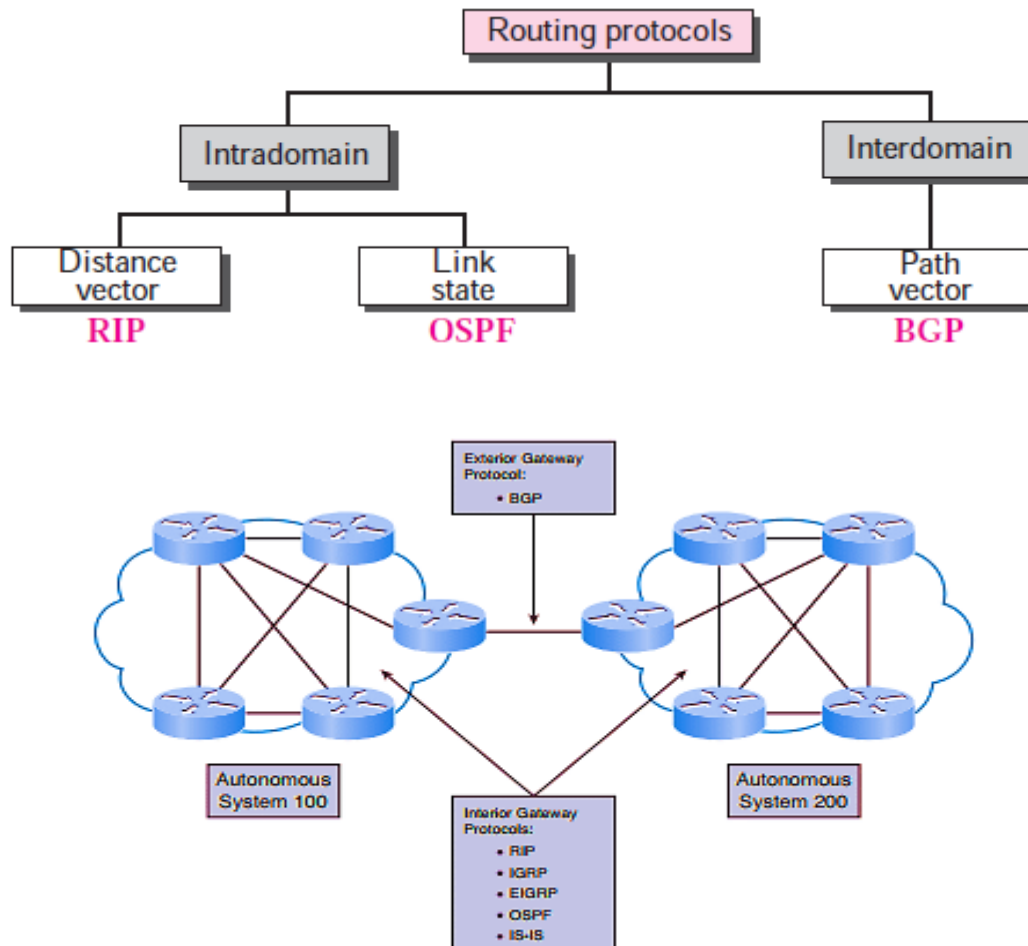
4.3- Routing Protocols

- A **routing protocol** is a combination of **rules and a procedure that lets** routers in the internet inform each other of **changes**.
- **Routing metric**: a method by which routing algorithms determines that one route is better than another route, Metric may be (*hop count, bandwidth, delay, or load*)

Interior Versus Exterior Routing Protocols

An autonomous system (AS), also known as a domain, is a **collection of routers** that are **under a common administration**, such as a company's internal network or an Internet service provider's (ISP's) network.

Because the Internet is based on the AS concept, two types of routing protocols are required:



4.4- Distance Vector Routing

- Distance Vector routing is **intra-domain** protocols, **inside** Autonomous system, but not between Autonomous system.
- distance-vector routing is based on the **least-cost** goal.
- Distance Vector developed by **Bellman-Ford** algorithm.
- Bellman **equation** is used to find the **least cost (shortest distance)** between a source to destination.
- A **distance vector routing algorithm** operates by having each router **maintain a table** (i.e., a **vector**) giving the best-known distance to each destination.
- These tables are **updated by exchanging** information with the neighbor's router. Every router knows the **best link** to reach each destination.
- **RIP** based on distance vector routing, each router **shares, at regular intervals**, its knowledge about entire AS with its neighbor.
- It is so **slow** and does not take **Bandwidth** into consideration when choose the root.

Distance vector algorithm

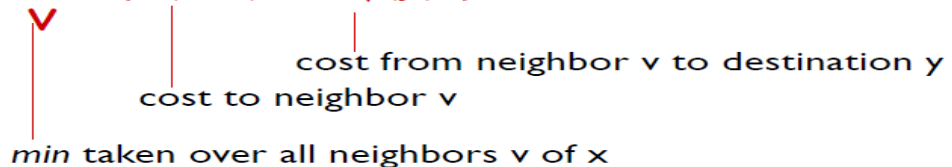
Bellman-Ford equation (dynamic programming)

let

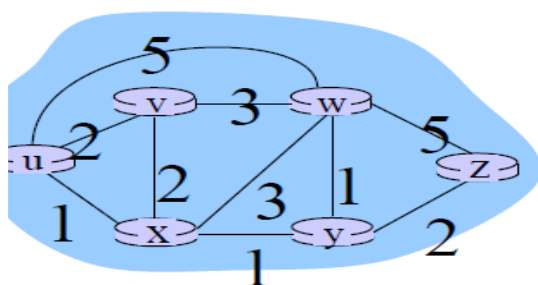
$d_x(y) := \text{cost of least-cost path from } x \text{ to } y$

Then

$$d_x(y) = \min \{ c(x,v) + d_v(y) \}$$



Bellman-Ford example



clearly, $d_v(z) = 5$, $d_x(z) = 3$, $d_w(z) = 3$

B-F equation says:

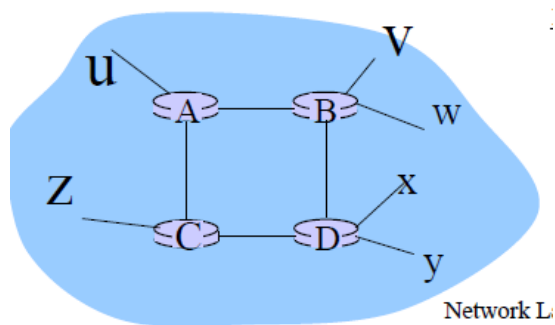
$$\begin{aligned}
 d_u(z) &= \min \{ c(u,v) + d_v(z), \\
 &\quad c(u,x) + d_x(z), \\
 &\quad c(u,w) + d_w(z) \} \\
 &= \min \{ 2 + 5, \\
 &\quad 1 + 3, \\
 &\quad 5 + 3 \} = 4
 \end{aligned}$$

node achieving minimum is next hop in shortest path, used in forwarding table

RIP (Routing Information Protocol)

- RIP is a routing protocol for exchanging routing table information between routers.
- It is a very simple protocol based on distance vector routing (DVR).
- simple intra-domain protocol
- Uses hop count as a path selection metric. (RIP prevents routing loops by implementing a limit on the number of hops allowed in a path from the source to a destination. *The maximum number of hops allowed for RIP is 15.* This hop limit, however, also limits the size of networks that RIP can support, a hop count of 16 is considered an **infinite distance**, in other words the route is considered unreachable.
- Three types of timers (**Routing-update timer (periodic timer) equal 30 seconds**, **Route timeout (Expiration Timer)**, if no update is received within the allotted 180 s, the route is considered expired and the hop count of the route is set to 16, which means the **destination is unreachable**, **Route-flush timer (Garbage Collection Timer)**, After the route timeout expires (240 s), the route-flush timer eventually expires, deleting the route from the table)

Example: Find router A routing table for the network topology shown below:



Dest. Net.	Next hop	Metric (Hop Count)
U	-	0
V	B	1
W	B	1
X	B	2
Y	B	2
Z	C	1

4.5- Link State Routing

The **primary problem in distance vector** that the algorithm often took **too long to converge** after the network topology changed (due to the count-to-infinity **problem**). Consequently, it was **replaced by** a new algorithm, now called link state routing.

The idea behind link state routing is simple and can be stated as five parts. Each router must:

- 1) **Discover** its neighbors and learn their network addresses.
- 2) **Measure** the delay or cost to each of its neighbors.
- 3) **Construct** a packet telling to all it has just learned.
- 4) **Send** the packet to all other routers.
- 5) **Compute** the shortest path to every other router.

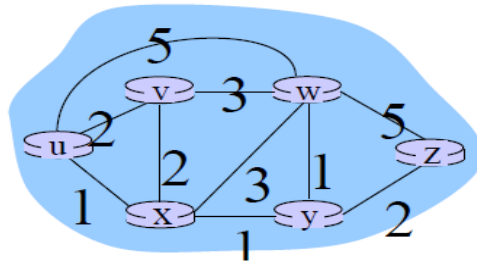
Compared to distance vector routing, link state routing requires **more memory and computation**. Also, the **computation time** grows faster

Nevertheless, in many practical situations, link state routing works **well because it does not suffer from slow convergence problems**.

Q: Consider the following network, define OSPF protocol & construct shortest path tree using Dijkstra algorithms

Dijkstra's algorithm: example

Step	N'	D(v),p(v)	D(w),p(w)	D(x),p(x)	D(y),p(y)	D(z),p(z)
0	u	2,u	5,u	1,u	∞	∞
1	ux	2,u	4,x		2,x	∞
2	uxy	2,u	3,y			4,y
3	uxyv		3,y			4,y
4	uxyvw					4,y
5	uxyvwz					



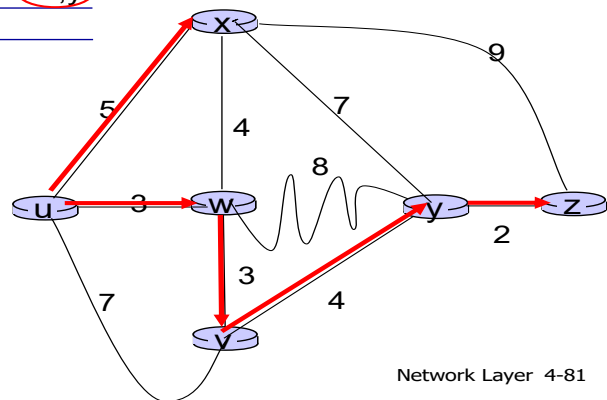
Q: Consider the following network, define OSPF protocol & construct shortest path tree using Dijkstra algorithms

Dijkstra's algorithm: example

Step	N'	D(v) p(v)	D(w) p(w)	D(x) p(x)	D(y) p(y)	D(z) p(z)
0	u	7,u	3,u	5,u	∞	∞
1	uw	6,w		5,u	11,w	∞
2	uwx	6,w			11,w	14,x
3	uwxv				10,y	14,x
4	uwxvy					12,y
5	uwxvyz					

notes:

- ❖ construct shortest path tree by tracing predecessor nodes
- ❖ ties can exist (can be broken arbitrarily)



Network Layer 4-81

Q: Compare between LSR & DVR

LSR	DVR
• All routers have complete topology, link cost info. • Updates are incremental and entire routing table is not sent as update. • Convergence is fast because of triggered updates. • More difficult to configure.	• Router knows physically-connected neighbors, link costs to neighbors • Entire routing table is sent as an update • Slow Convergence due to the count to infinity problem. • Easy to configure.

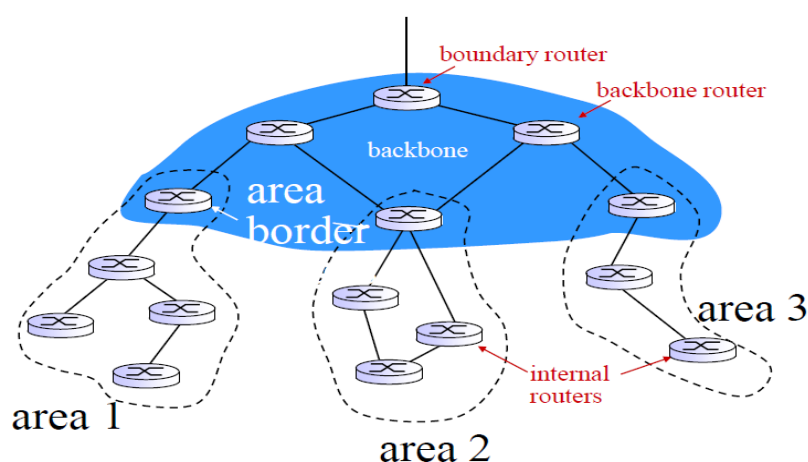
OSPF (Open Shortest Path First)

It uses a **link state routing algorithm** and falls into the group of interior routing protocols, operating within a single autonomous system (AS).

- **Topology map** at each node
- route computation using **Dijkstra's algorithm**
- OSPF is perhaps the most widely **used interior gateway protocol (IGP)** in large enterprise networks.
- The metric of OSPF is the **cost** of sending packets across a certain interface.
- **"open"**: publicly **available**
- **"Security"**: all OSPF messages authenticated (to **prevent malicious intrusion**)
- Hierarchical OSPF in large domains.

Hierarchical OSPF

- *Two-level hierarchy: local area, backbone.*
- Link-state advertisements only in **area**
- Each node has **detailed area topology**; only know direction (shortest path) to nets in other areas.
- **Area border routers**: “**summarize**” distances to nets in own area, advertise to other Area Border routers.
- **Backbone routers**: run OSPF routing limited to backbone.
- **Boundary routers**: connect to other AS's.



OSPF is superior to RIP in all aspects, including the following:

- It converges much faster.
- It supports hierarchical structures.
- It has improved metric calculation for best path selection.
- It does not have hop-count limitations
- At its inception, OSPF supported the largest networks.
- Compare to RIP, OSPF has no limitation due to hops (RIP has a limit of 15 hops so any network with more than 15 hops cannot be achieved by RIP).

4.6- Path Vector Routing

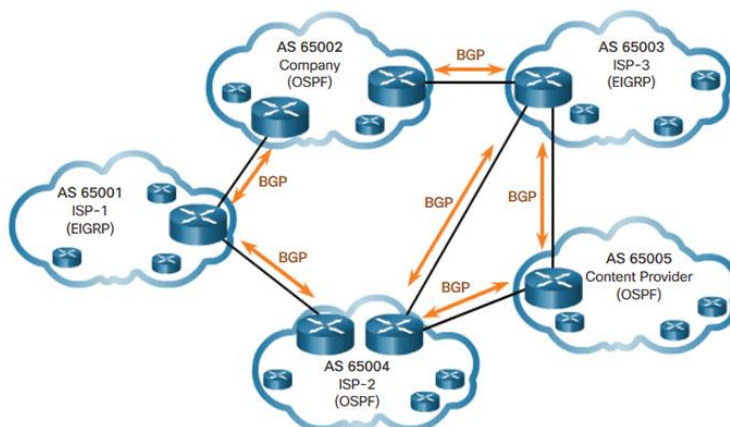
Path vector routing is an interdomain, policy-based protocol where routers advertise the entire path (a sequence of autonomous systems) to a destination, rather than just the distance/next-hop. It prevents routing loops by detecting if the local AS is already in the path, primarily used in the Border Gateway Protocol (BGP).

Key Aspects of Path Vector Routing

- **Loop Prevention:** Because the path vector contains the list of Autonomous Systems (AS) a packet will traverse, a router can instantly detect if a route loops back to its own AS and discard it.
- **Policy-Based:** Rather than just finding the shortest path, routers use policies (like cost, speed, or security) to select the best route, making it ideal for internet routing between different organizations.
- **Speaker Node:** In each AS, one specialized node ("speaker") advertises reaching network information to neighboring speaker nodes in other ASs.
- **Example (BGP):** The Border Gateway Protocol is the standard protocol for interdomain routing on the internet, using path vectors to ensure loop-free, policy-compliant path selection.

Comparison with Other Protocols

- **Vs. Distance Vector:** Distance vector only shares the next hop and distance (metric), leading to the "count to infinity" problem, whereas path vector eliminates this by listing the *full path*.
- **Vs. Link State:** Path vector is *more scalable* for large, interdomain networks and focuses on policy, while link state (like OSPF) provides faster convergence within a single domain.

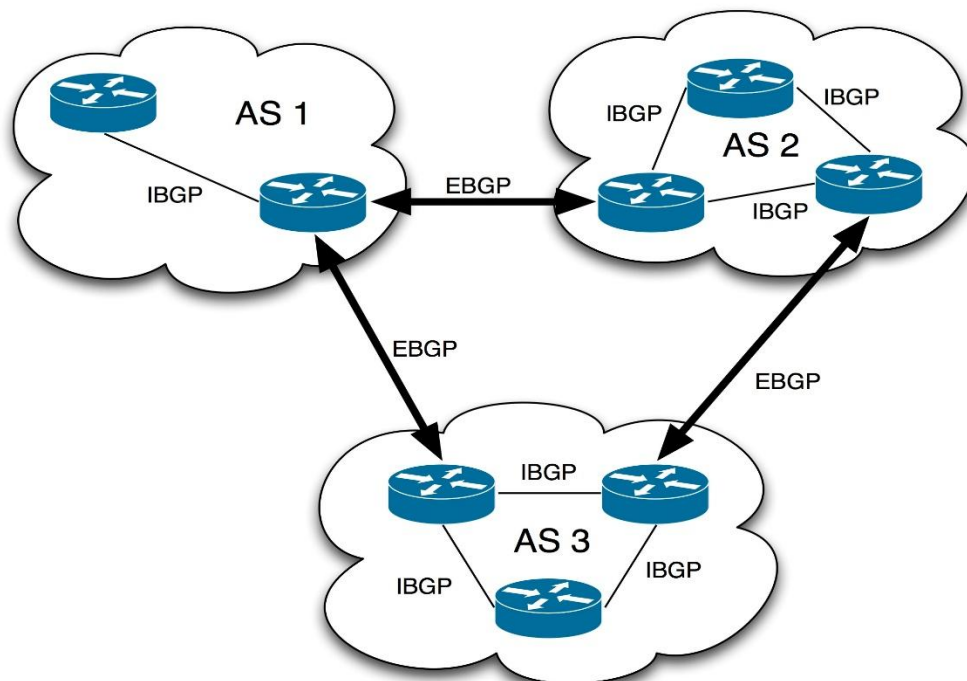


Border Gateway Protocol (BGP)

- **Inter-domain routing protocol** for routing between autonomous systems (holds the Internet together)
- BGP is neither a link state, nor a distance vector protocol, it is known as **path vector protocol**. Routing messages in BGP contain complete routes.
- Network administrators can specify routing policies (**BGP supports flexibility** - paths could be chosen by a provider based on a policy).
- Network administrators can specify routing policies.
- BGP's goal is to find **any path** (not an optimal one).

Internal and external BGP sessions

iBGP	eBGP
• Used to connect different routers have same AS (same company) • Propagate reachability information to all AS-internal routers.	• Used to connect different routers have different AS (different company) • Obtain subnet reachability information from neighboring ASs.





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Computer Network Protocols

Chapter Five

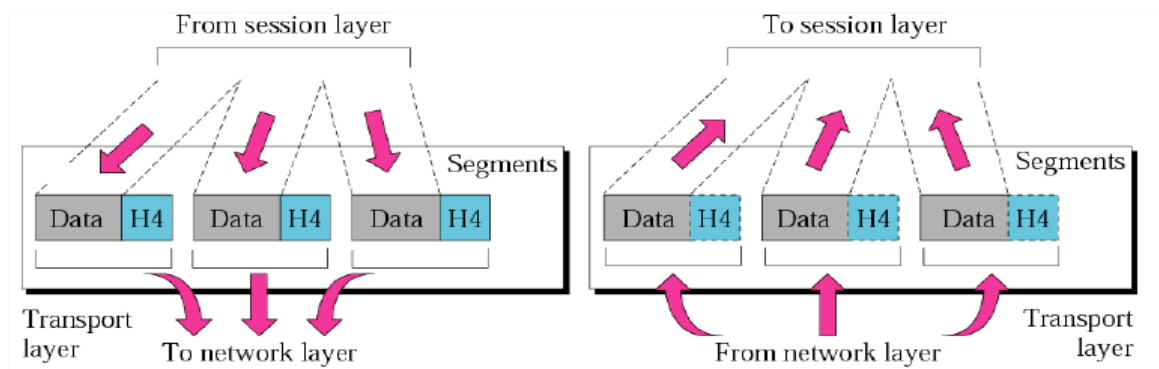
Transport & Application Layer Protocols

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5.1- Transport Layer Functions

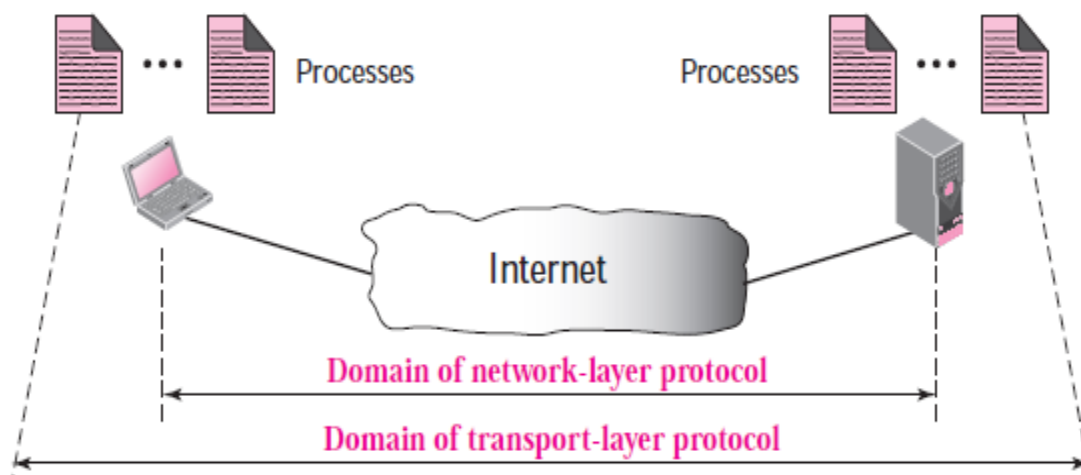
The main functions of transport layer are:

- End to end data delivery.
- Receives data from upper layers and segments it into packets.
- Can provide error correction and recovery.
- End to end flow control.
- Congestion control.
- Quality of Service (QoS).
- Connection oriented and connection less.



Transport layer provides **logical communication** between application processes running on different hosts, Transport protocols run in **end systems**

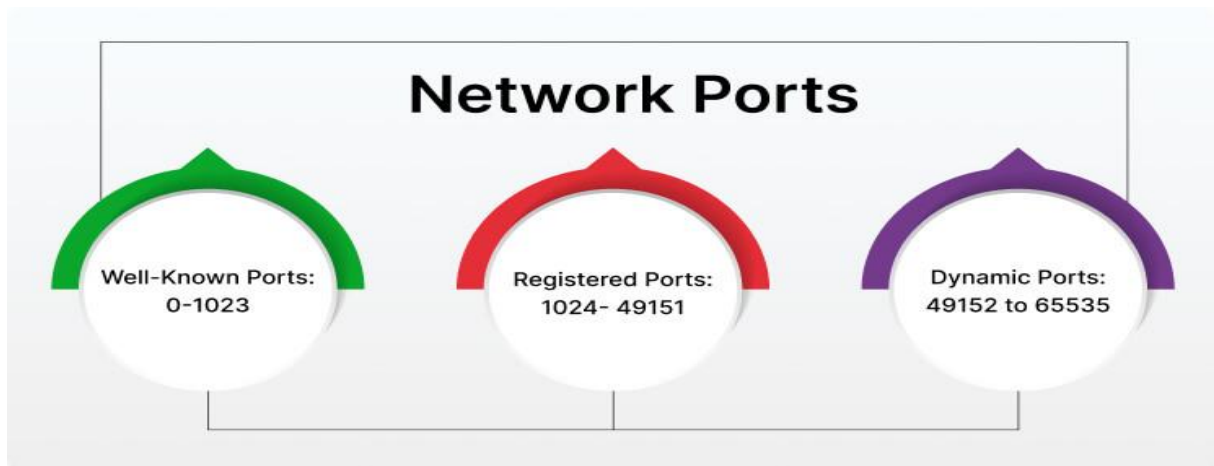
- **Network layer:** logical communication **between hosts**
- **Transport layer:** logical communication **between processes**



5.2- Port Address

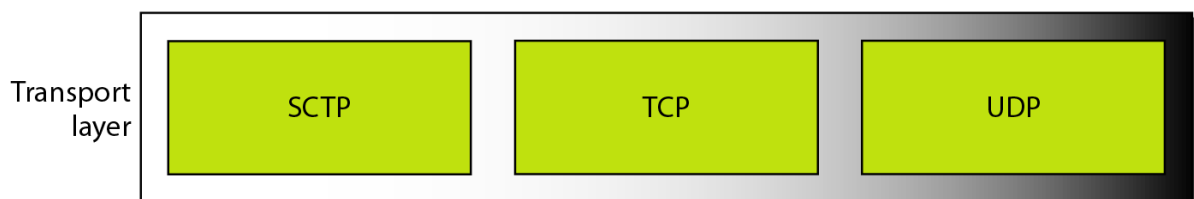
The **IANA** (Internet Assigned Number Authority) has divided the port numbers into three ranges: *well-known*, *registered*, and *dynamic (or private)*:

- **Well-known ports:** The ports ranging from **0 to 1023** are assigned and controlled by IANA. These are the well-known ports.
- **Registered ports:** The ports ranging from **1024 to 49,151** are not assigned or controlled by IANA. They can only be registered with IANA to prevent duplication.
- **Dynamic ports:** The ports ranging from **49,152 to 65,535** are neither controlled nor registered. They can be used by any process. These are the ephemeral ports.



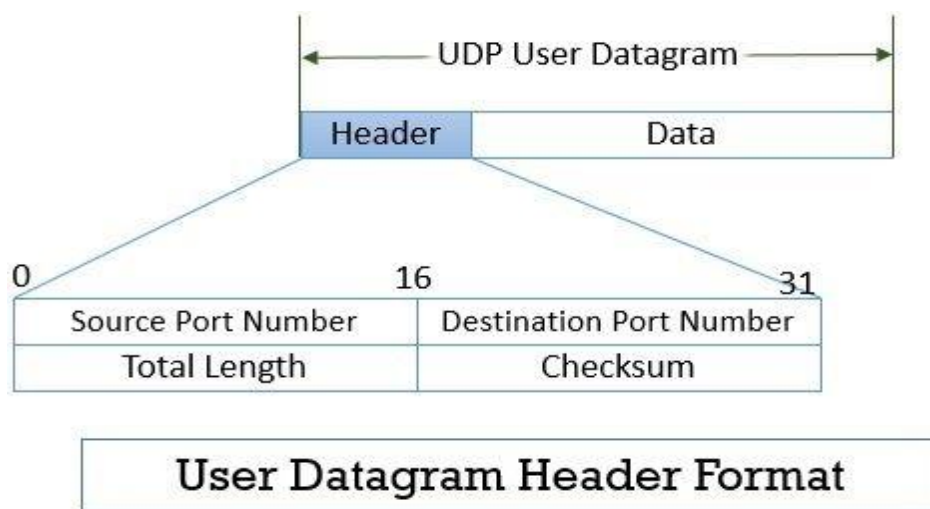
5.3- The Transport Layer Protocols:

The main protocols of the Transport layer are the **UPD**, **TCP**, and **SCTP** protocols.



5.3.1- User Datagram Protocol (UDP)

- **Connectionless (unreliable transport protocol)** (best-effort)
- This means that each user datagram sent by UDP is an **independent datagram** and no handshaking between UDP sender, receiver.
- UDP segments may be: (lost, delivered **out-of-order** to app)
- The user datagrams **are not numbered**.
- Also, there is **no connection establishment and no connection termination**, this means that each user datagram **can travel on a different path**.
- UDP is a very **simple protocol using a minimum of overhead**.
- There is **no flow control** and hence **no window mechanism**.
- Unreliable but **fast**



Q: Why is there a UDP?

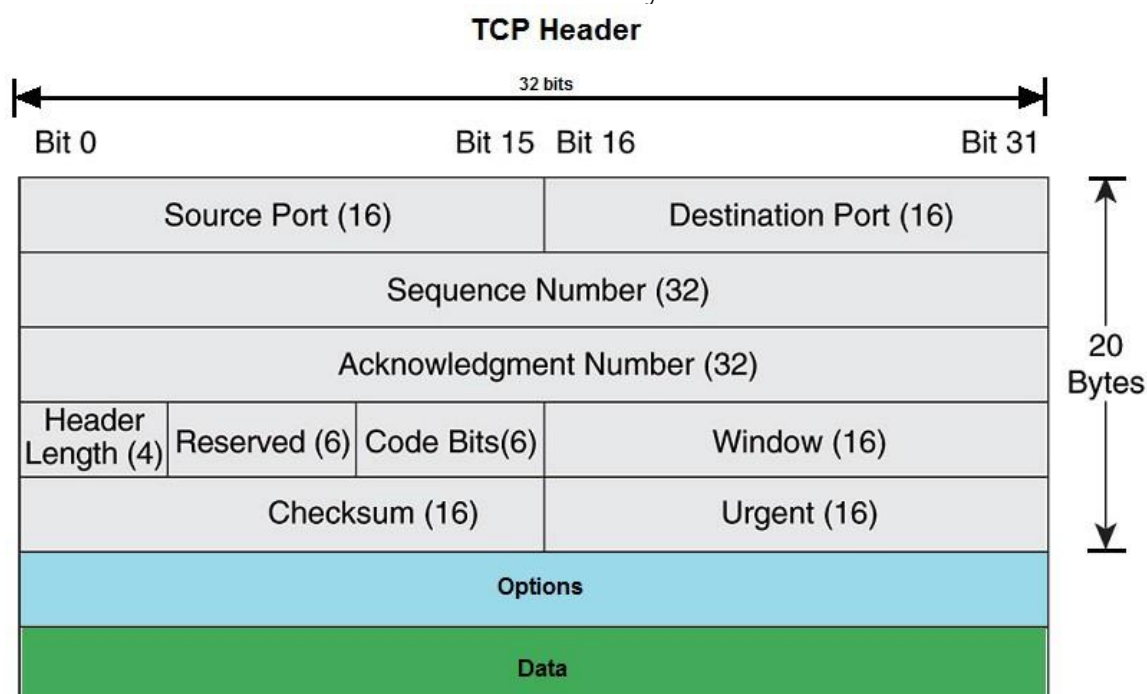
- No connection establishment (which can add delay)
- Simple: no connection state at sender, receiver
- Small header size
- No congestion controls.

The following lists some uses of the UDP protocol:

- Routing Information Protocol (**RIP**)
- Domain Name System (**DNS**)
- Streaming Multimedia Applications (loss tolerant, rate sensitive)
- Simple Network Management Protocol (**SNMP**)

5.3.2- Transmission Control Protocol (TCP)

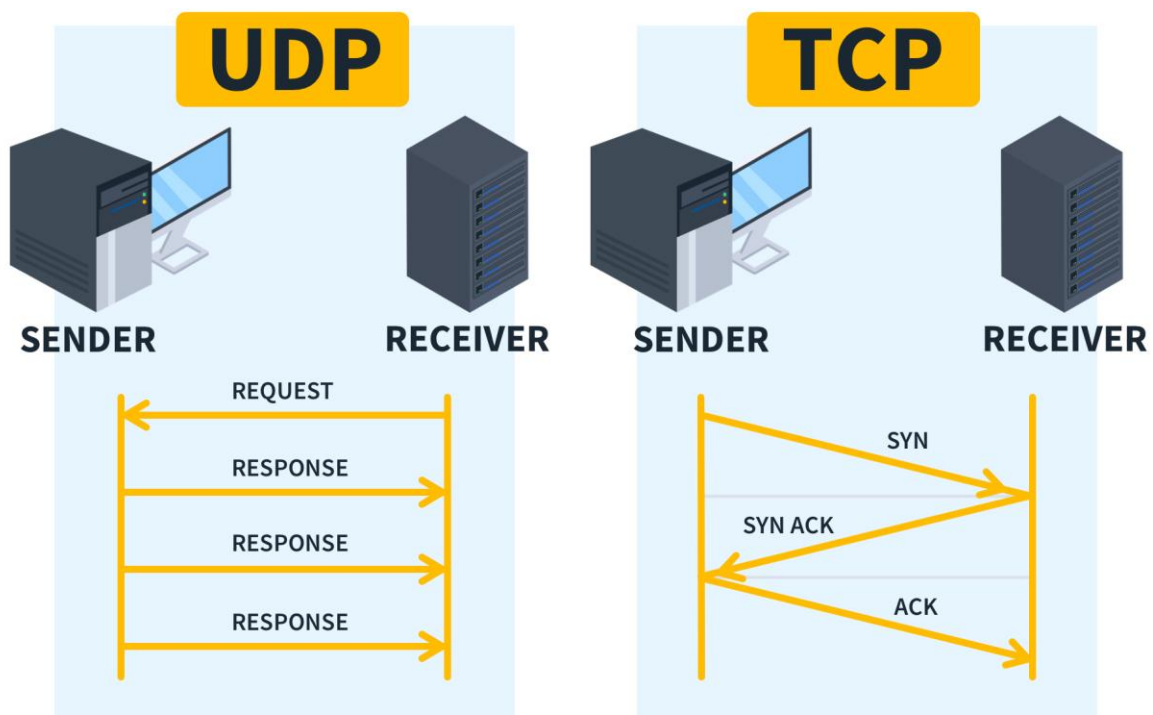
- Reliable, in-order delivery (congestion control, flow control, connection setup)
- TCP, **like** UDP, is a **process-to-process** (program-to-program) protocol. TCP, therefore, like UDP, **uses port numbers**.
- **Unlike** UDP, TCP is a **connection-oriented** protocol; it creates a virtual connection between two TCPs to send data.
- In addition, TCP **uses flow and error control mechanisms** at the transport level.
- **Reliable delivery**: no packet loss, error, duplication, disorder
- In brief, TCP is called a **connection-oriented, reliable** transport protocol. It adds connection-oriented and reliability features to the services of IP.



Q: Compare between UDP and TCP protocols

UDP	TCP
1. Connectionless service, UDP datagrams are delivered independently. 2. Unreliable delivery: packet loss, corruption, duplication, disorder.	1. Connection-oriented: connection establishment & termination 2. Reliable delivery: no loss, error, duplication, disorder.

3. Fast as compared with TCP 4. No sequencing 5. No acknowledge 6. Simple request-response communication without internal flow and error control 7. RIP, DNS use UDP	3. Slow: retransmission, flow control 4. Segment sequencing 5. Acknowledge segment 6. Connection overhead 7. Telnet, SMTP, HTTP and FTP, use TCP
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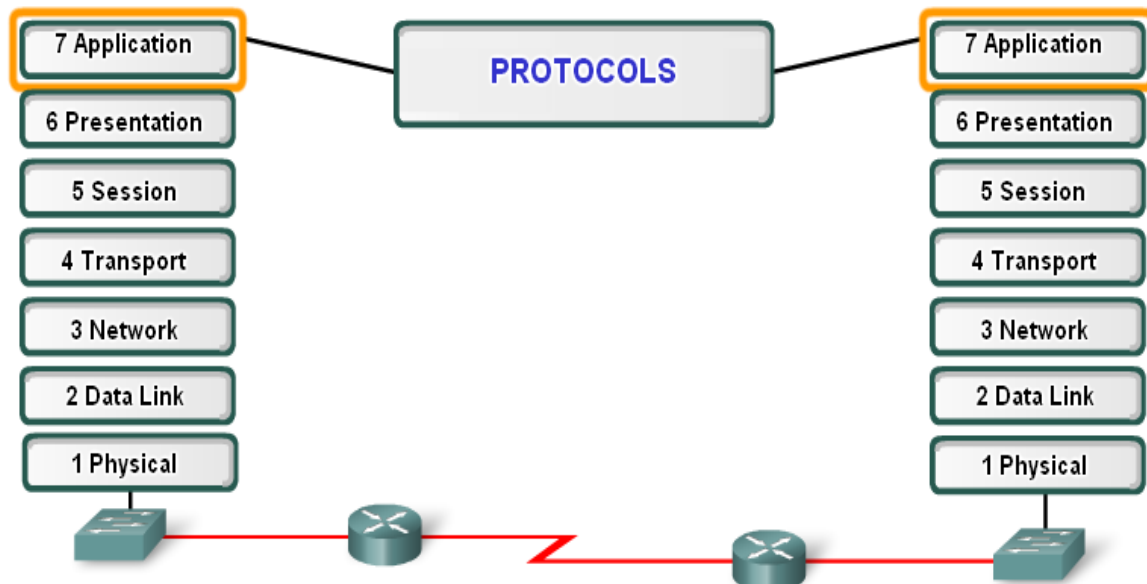


5.3.3- SCTP (Stream Control Transmission Protocol):

- ✓ it is a reliable message-oriented protocol.
- ✓ combines the best features of UDP and TCP. SCTP is a reliable message-oriented protocol.
- ✓ Flow and error control
- ✓ Congestion control

5.4- Application layer Protocols

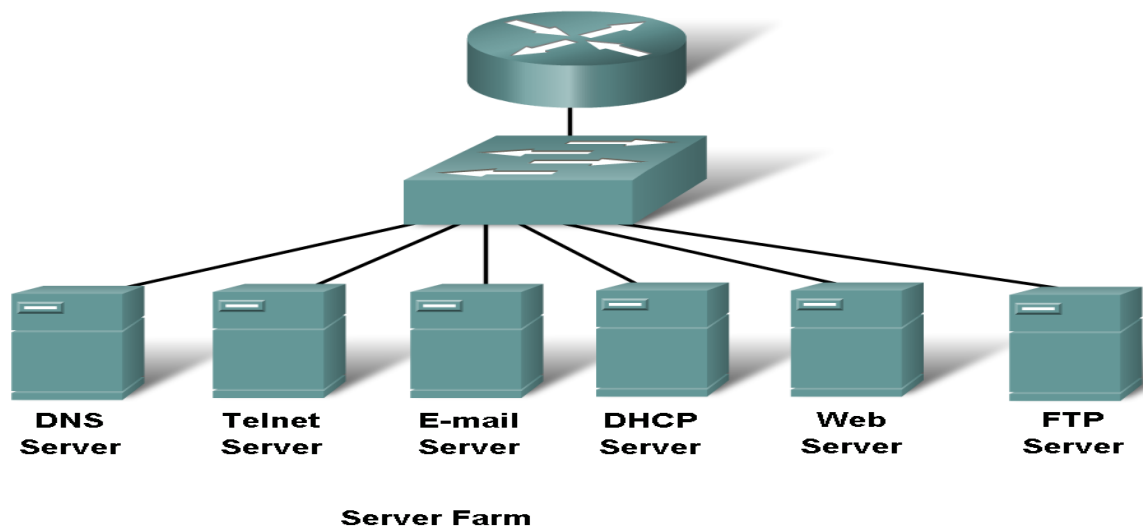
- Allows user to interface with the network.
- Provides the interface between the applications on either end of the network.



Application layer protocols provide the rules for communication between applications.

Protocols:

- Define processes on either end of the communication
- Define the types of messages
- Define the syntax of messages
- Define the meaning of any informational fields
- Define how messages are sent and the expected response
- Define interaction with the next lower layer

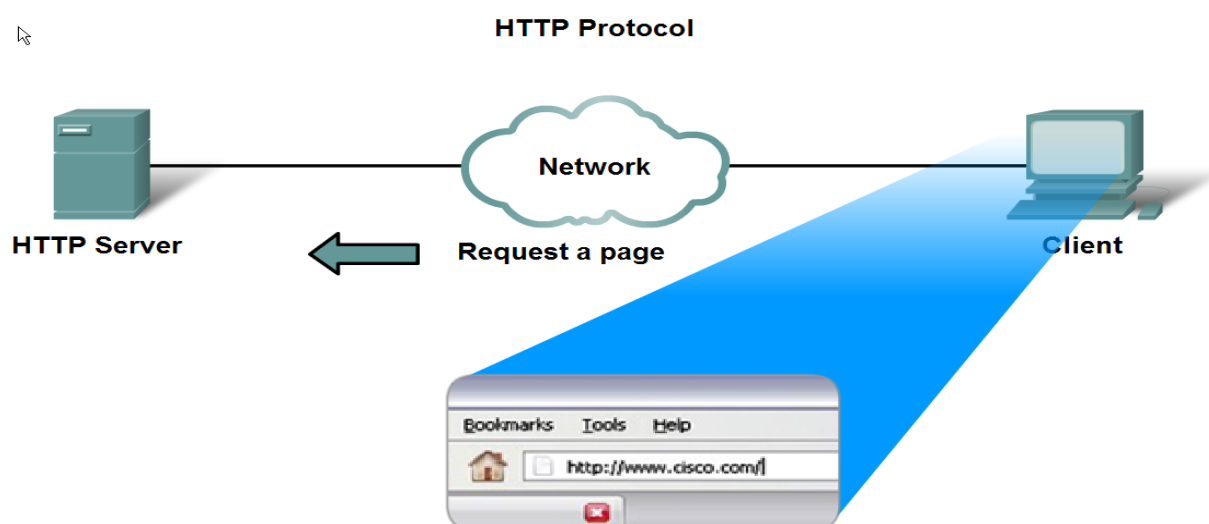


Protocols	Description
DNS	Matches domain names with IP addresses
HTTP	Used to transfer data between clients/servers using a web browser
SMTP & POP3	used to send email messages from clients to servers over the internet
FTP	allows the download/upload of files between a client/server
Telnet	allows users to login to a host from a remote location and take control as if they were sitting at the machine (virtual connection)
DHCP	assigns IP addresses, subnet masks, default gateways, DNS servers, etc. To users as they login the network

Common Port Numbers

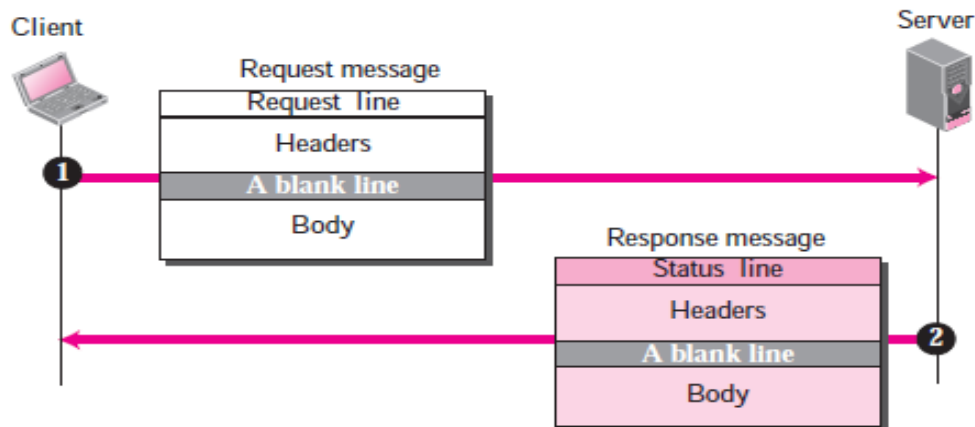
TCP	UDP
• FTP – 20-21 • Telnet – 23 • SMTP – 25 • DNS – 53 (Both TCP & UDP) • HTTP – 80	• DHCP – 67 & 68 • POP – 110

5.4.1- Hypertext Transfer Protocol (HTTP)



- The (HTTP) is a protocol used mainly to access data on the World Wide Web.
- HTTP use TCP connection and **port 80**.

- HTTP messages are not destined to be read by humans; (read and interpreted by the HTTP server and HTTP client (browser)).
- It is a **stateless protocol**, which means that the server does not keep information about the client. The client initializes the transaction by sending a request. The server replies by sending a response.



- A client that wants to access a Web page needs the **file name and the address**.
- The **uniform resource locator (URL)** is a standard locator for specifying any kind of information on the Internet, The URL defines four things:

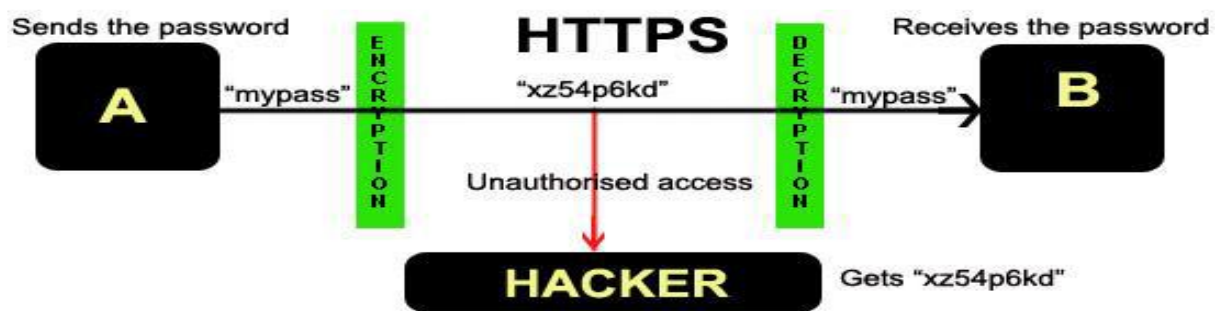


Steps:

- 1- URL is typed in the address bar.
- 2- Browser checks with DNS server to convert it to an IP address
- 3- Connects to the server requested
- 4- Using HTTP or HTTPS protocol requirements, the browser sends a GET request to the server to ask for the desired html document (usually index.html)
- 5- The server sends the HTML code for the web page to the browser.
- 6- The browser interprets the HTML code and formats the page to fit the browser window.

HTTPS (HTTP + SSL)

- **HTTPS** = HTTPS stands for Hypertext Transfer Protocol over **Secure Socket Layer** (SSL), or HTTP over SSL.
- HTTPS by default uses port 443.
- URL's beginning with HTTPS indicate that the connection is encrypted using SSL.
- The main **disadvantage of HTTPS** is that it is slightly slower than HTTP because of the encryption of data.

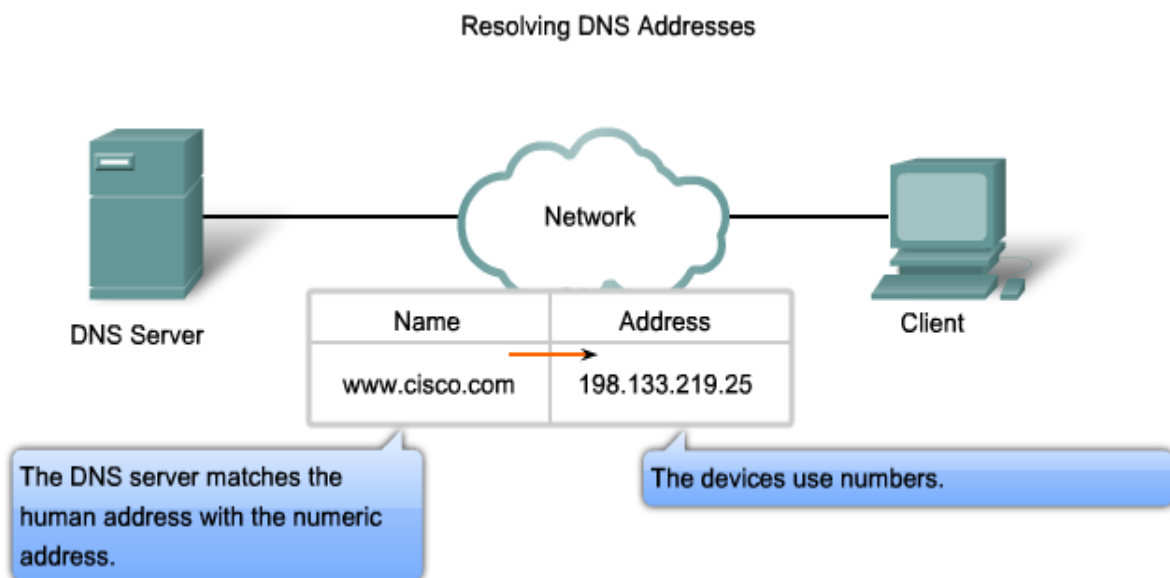


Q: Compare between HTTP & HTTPS.

HTTP	HTTPS
1. URL begins with "http://"	1. URL begins with "https://"
2. unsecured	2. secured
3. uses port 80 for communication	3. uses port 443 for communication.
4. operates at Application Layer	4. operates at Transport and application Layer.
5. No encryption	5. uses encryption
6. No certificates required	6. certificates required

5.4.2- Domain Name Service (DNS)

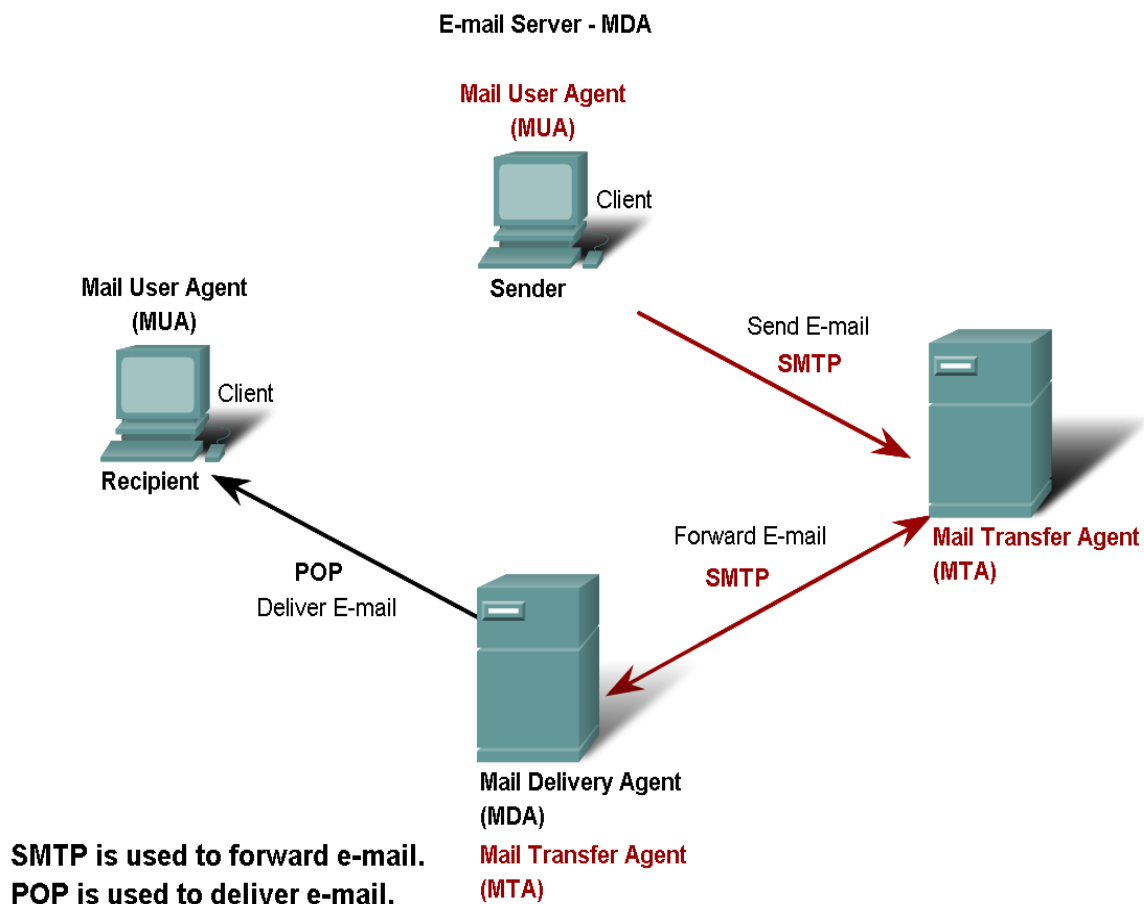
- It is a client-server application that identifies each host on the Internet with a **unique user-friendly name**.
- The names must be unique because the addresses are unique.
- DNS Servers resolve names to IP addresses. It would be difficult to remember the IP address of every website we like to visit, but we can remember names.



Record Type	Purpose
Addresses	
A	Maps a fully qualified domain name (FQDN) to an IPv4 address
AAAA	Maps a FQDN to an IPv6 address
Aliases	
CNAME	Maps a FQDN to another FQDN
DNAME	Maps all subdomains of a FQDN to another FQDN
Servers	
NS	Maps a subdomain to a FQDN of a name server
MX	Maps an email domain to a FQDN of a mail server

5.4.3- E-mail Protocols (SMTP/POP/IMAP)

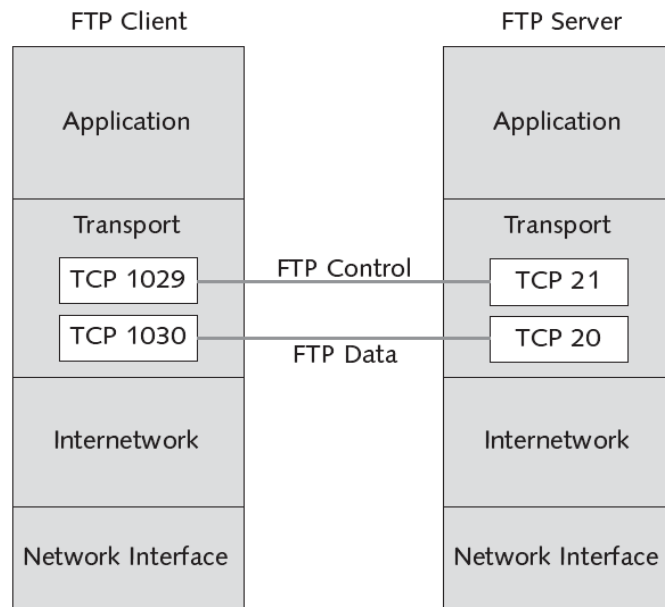
- E-mail is the most popular network **service**.
- E-mail **client** (when people compose e-mail) is called Mail User Agent (**MUA**)
- MUA **allows** messages to be **sent/retrieved** to and from your mailbox
- Requires several applications and services:
 - ✓ **POP or POP3** – deliver email from server to client (incoming messages) post office port.
 - ✓ **SMTP** – handles outbound messages from clients



Q- What is the difference between POP3 and IMAP protocols?

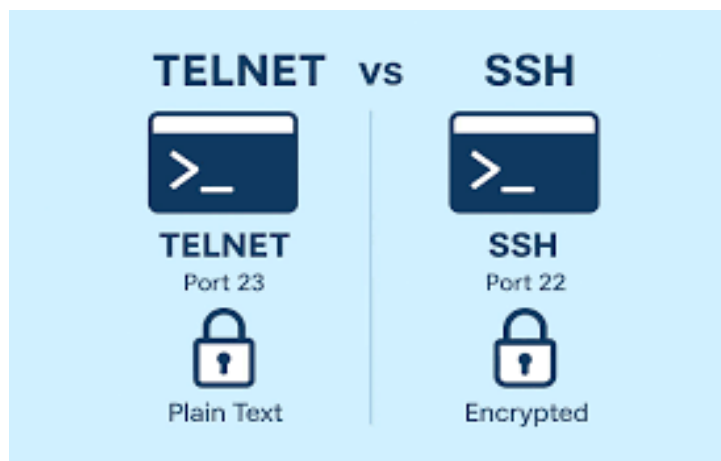
5.4.4- File Transfer Protocol (FTP)

- Commonly used application layer protocol
- Allows for the **transfer of files** between clients/servers.
- Requires 2 connections to the server:
Commands – uses TCP port 21
Actual data – uses TCP port 20



5.4.5- TELNET Protocol

- TELNET is an abbreviation for **TErminaL NETwork**.
- It is a protocol for virtual terminal service, enables the establishment of a connection to a remote system in such a way that the local terminal appears to be a terminal at the remote system.
- It uses port 23.
- The secure version of telnet is SSH (Secure Shell) protocol, which encrypts all traffic on port 22





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Chapter Six

The Data Link Layer Protocols

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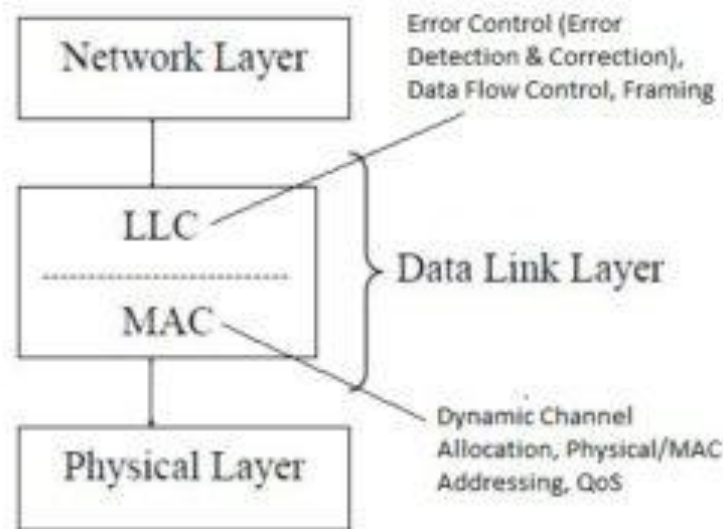
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6.1- Introduction

Data-link layer has the responsibility of transferring datagram from one **node to physically adjacent node** over a link. The data link layer is divided into two sublayers: **Logical Link Control (LLC)** and **Media Access Control (MAC)**. The first, *logical link control, deals with the communication between two adjacent nodes: node-to-node communication. The second function of the data link layer is media access control, or how to share the link.*

Logic link control functions include:

- ***Framing***
- ***Flow Control***
- ***Error control.***



6.2- Framing

Framing in the data link layer separates a message from one source to a destination, by adding a sender address and a destination address. Although the whole message could be packed in one frame; this is not normally done. One reason is that a frame can be very large, making flow and error control very inefficient.

6.3- Flow and Error Control

The most important responsibilities of the data link layer are *flow control* and *error control*. Collectively, these functions are known as **data link control**.

6.3.1- Flow Control

Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before receiving an acknowledgment and is one of the most important duties of the data link layer. The flow of data must **not be allowed to overwhelm the receiver**. Any receiving device has a limited speed at which it can *process* incoming data and a limited amount of memory in which to *store* incoming data. *The receiving device must be able to inform the sending device before those limits are reached and to request that the transmitting device send fewer frames or stop temporarily.* Incoming data must be checked and processed before they can be used. *The rate of such processing is often slower than the rate of transmission.* For this reason, each receiving device has a block of memory, called a **buffer**, reserved for storing incoming data until they are processed. If the buffer begins to fill up, the receiver must be able to tell the sender to halt transmission until it is once again able to receive.

6.3.2- Error Control

Error control is both *error detection* and *error correction*. It allows the receiver to inform the sender of any frames lost or damaged in transmission and coordinates the retransmission of those frames by the sender. In the data link layer, the term error control refers primarily to methods of error detection and retransmission. Error control in the data link layer is often implemented simply. Any time an error is detected in an exchange, specified frames are retransmitted. This process is called ***automatic repeat request (ARQ)***.

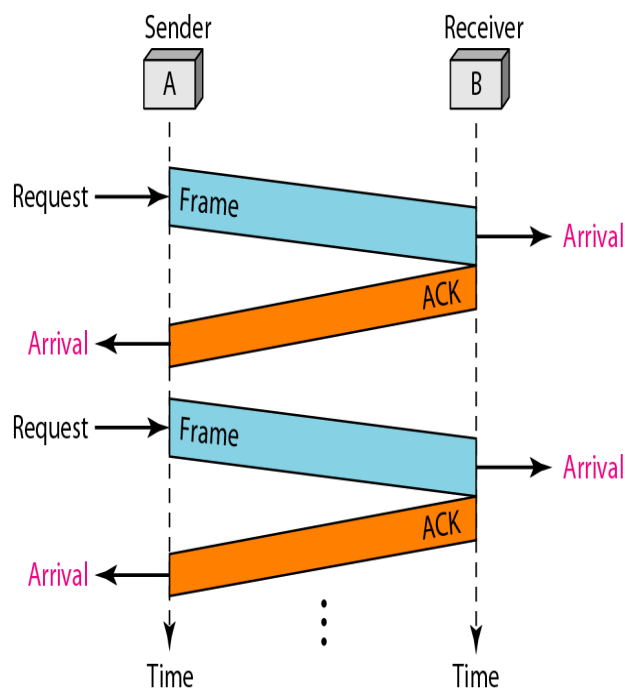
6.4- Protocols

Now let us see how the data link layer can combine framing, flow control, and error control to achieve the delivery of data from one node to another.

6.4.1- Stop-and-Wait Protocol

If data frames arrive at the receiver site *faster than they can be processed*, the frames must be stored until their use. To prevent the receiver from becoming overwhelmed with frames, we need to tell the sender to slow down. There must be feedback from the receiver to the sender.

The protocol is called the **Stop-and-Wait Protocol** because the sender sends one frame, stops until it receives confirmation from the receiver (okay to go ahead), and then sends the next frame.

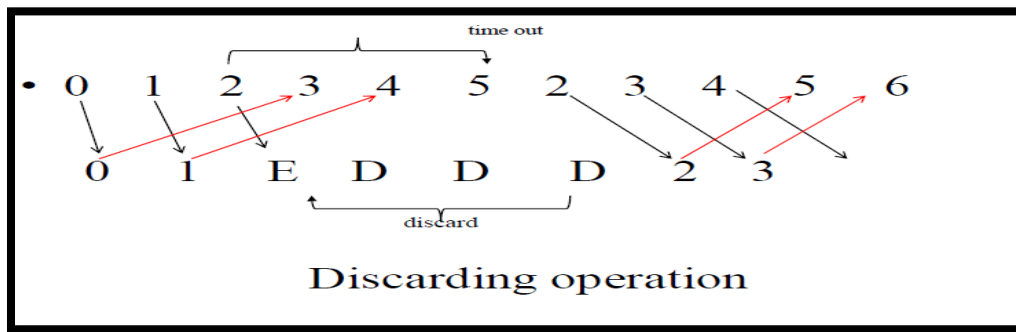


6.4.2- Sliding Window Protocols

To improve the efficiency of transmission, multiple frames must be in transition while waiting for acknowledgment. In other words, we need to let more than one frame be outstanding to keep the channel busy while the sender is waiting for acknowledgment. This can be done by using **Go-back-N** or **Selective-Repeat** protocols rather than Stop-and-Wait. These protocols used for flow and error control but at transport layer.

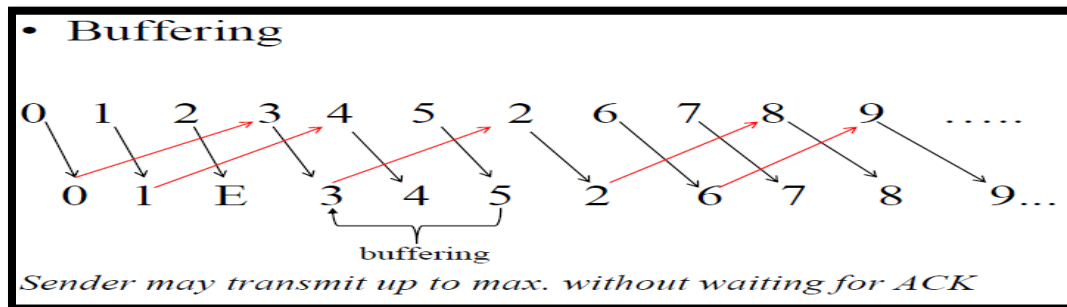
Go-back-N Protocol

If there is one frame k missing, the receiver simply discards all subsequent frames $k+1, k+2, \dots$, sending no acknowledgments. So, the sender will retransmit frames from k onwards. This can be a waste of bandwidth.



Selective-Repeat Protocol

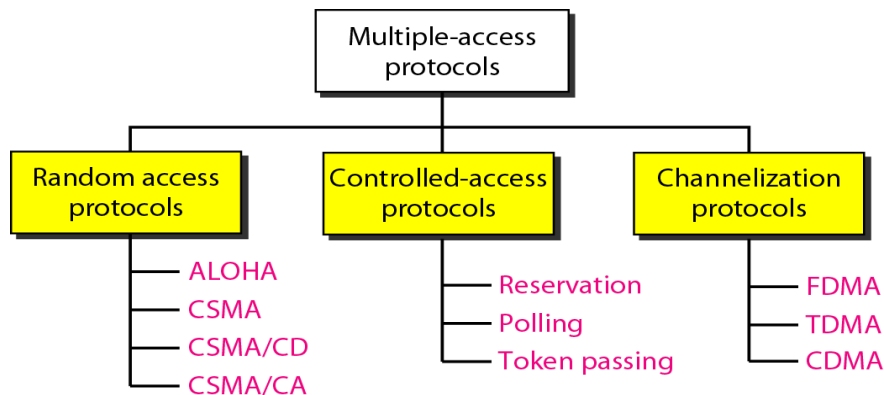
Another strategy is to re-send only the ones that are actually lost or damaged. The receiver buffers all the frames after the lost one. When the sender finally noticed the problem (e.g. no ack for the lost frame is received within time-out limit), the sender retransmits the frame in question.



6.5- Multiple Accesses Protocols

The second sub-layer of the data link layer as we said previously is the media access control. There are two main types of the media access control; **Point-to Point access** and **Multiple Access**.

When there is a *dedicated link can be that can be used by the two devices at any time*; this is known as *Point-to-Point Access*. While, when *the link is shared between pairs of devices that need to use the link*; this is known as a *Multiple Access*.



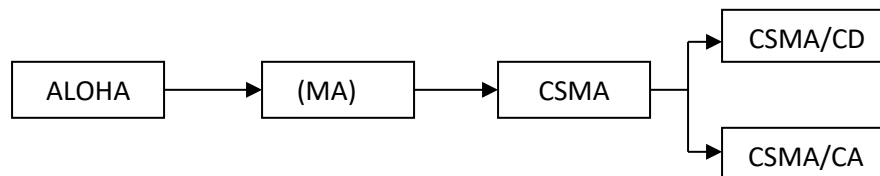
6.5.1- Random Access

In random access, *no station is superior to another station and none is assigned the control over another*. No station permits, or does not permit, another station to send. There is no scheduled time for a station to transmit. Transmission is random among the stations. That is why these methods are called *random access*. In a random-access method, each station has the right to the medium without being controlled by any other station. However, if more than one station tries to send, there is an access conflict-*collision*-and the frames will be either destroyed or modified.

The random-access methods have evolved from a very interesting protocol known as **ALOHA**, which used a very simple procedure called **Multiple Access (MA)**. The method was improved with the addition of a procedure that forces the station to sense

the medium before transmitting. This was called **Carrier Sense Multiple Access (CSMA)**.

This method later evolved into two parallel methods: **Carrier Sense Multiple Access with Collision Detection (CSMA/CD)** and **Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)**. The first tells the station what to do when a collision is detected, while the second tries to avoid the collision.

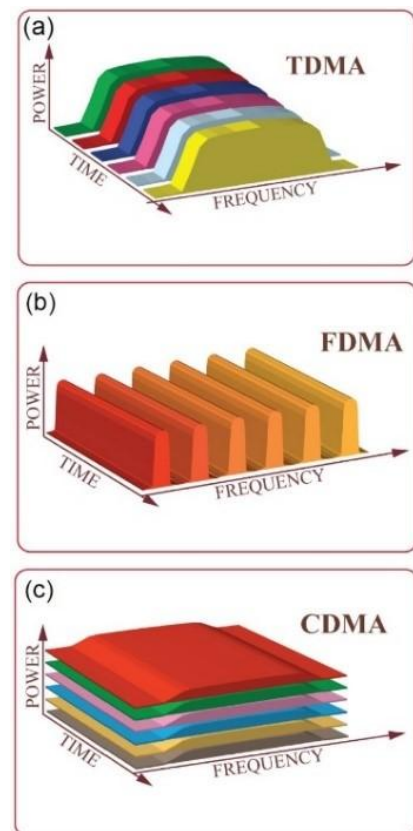


6.5.2- Controlled Access

In controlled access, the stations consult one another to find which station has the right to send. A station cannot send unless it has been authorized by other stations.

6.5.3- Channelization

Channelization is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations. Therefore, it is known as FDMA, TDMA, and CDMA. These methods are similar to the multiplexing methods but the main difference is that they occur at the data link layer and are used for multiple access for the shared link of the wireless networks.



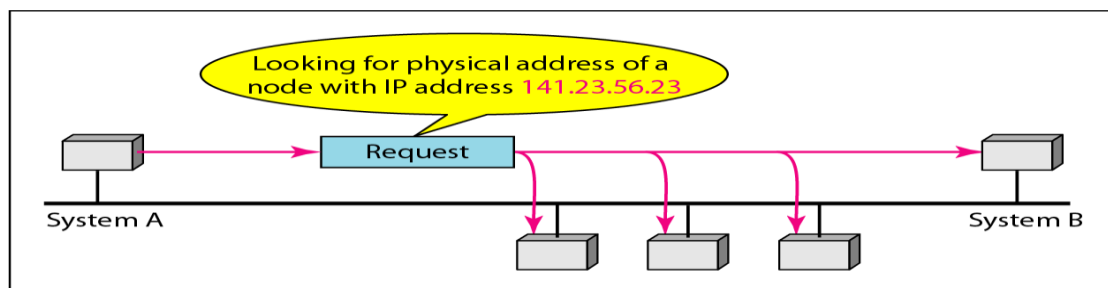
6.6- Layer 2 Protocols

6.6.1- Address Resolution Protocol (ARP)

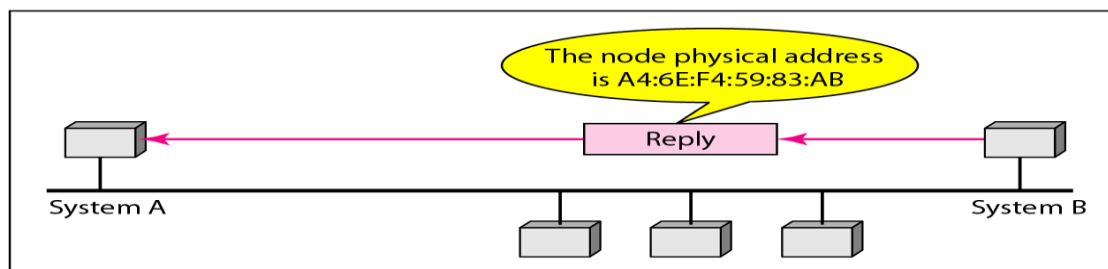
The delivery of a packet to a host or a router requires two levels of addressing: **logical** and **physical**.

- ARP Maps IP addresses to MAC addresses
- ARP **Request** is a broadcast but ARP **reply** is Unicast.
- **ARP tables** contain the MAC and IP addresses of other devices on the network

ARP operation:



a. ARP request is broadcast



b. ARP reply is unicast

6.6.2- Layer 2 Tunneling Protocol (L2TP)

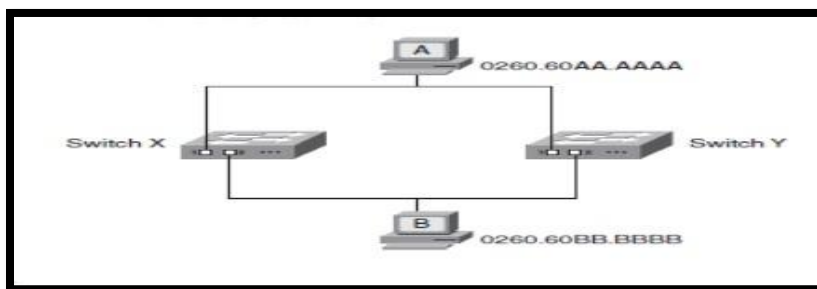
- Is an **extension** of the Point-to-Point Tunneling Protocol (PPTP).
- **Used by** an Internet service provider (ISP) to enable the operation of a **Virtual Private Network (VPN)** over the Internet.
- The goal of a Virtual Private Network (VPN) is to **provide private communications within the public Internet Infrastructure**

6.6.3- Spanning Tree Protocol (STP)

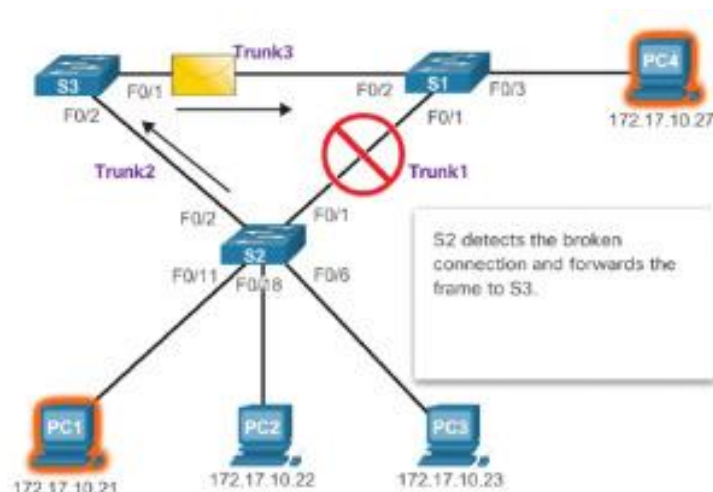
Spanning Tree Protocol (STP) was developed **to prevent the broadcast storms caused by switching loops**, it is a Layer 2 protocol that runs on **bridges and switches**. The main purpose of STP is to ensure that you do not create loops when you have redundant paths in your network.

Switching Loops

Redundancy in a network, such as that shown in Figure below, is **desirable** so that communication can still take place if a link or device fails. **For example**, if switch X in this figure stopped functioning, devices A and B could still communicate through switch Y. However, in a switched network, **redundancy can cause problems which is called switching loop**.



When a switching loop is introduced into the network, a destructive **broadcast storm** will develop **within seconds**. To overcome these problems, you must have a way to logically disable part of the redundant network for regular traffic while maintaining redundancy for the case when an error occurs. Spanning Tree Protocol (STP) does just that.



6.6.4 HDLC Protocol

High-level Data Link Control (**HDLC**) is a bit-oriented layer 2 protocol for communication over point-to-point and multipoint links. It implements the ARQ mechanisms. The primary purpose of **HDLC** is to ensure reliable, error-free, and orderly transmission of data frames between network nodes over a synchronous serial communication link.

6.6.5- Point to Point Protocol (PPP)

Although HDLC is a general protocol that can be used for both point-to-point and multipoint configurations, one of the most common protocols for point-to-point access is the Point-to-Point Protocol (PPP). Today, millions of Internet users who need to connect their home computers to the server of an Internet Service Provider (ISP) use PPP.

Q: Compare between HDLC and PPP protocols

HDLC	PPP
Stand for High-level Data Link Control	Stand for Point-to-Point Protocol
Bit-oriented protocol	Byte-oriented protocol
Support Point to Point and Multipoint Links	Support Point to Point links Only
No Authentication Mechanism	There is Authentication in PPP
Flow Control	No Flow Control
Error Detection and retransmission for recovery	Error detection but there is no recovery mechanism

PPPoE

(Point-to-Point Protocol over Ethernet) is a network protocol that encapsulates PPP frames inside Ethernet frames, commonly used by Internet Service Providers (ISPs) to manage customer broadband connections (DSL, fiber, cable). It enables authentication, encryption, and compression using a username and password to establish a secure session between a home router and the ISP.